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Η έγκριση της Διπλωματικής Εργασίας από το Τμήμα Στατιστικής και Ασφαλιστικής Επιστήμης του Πανεπιστημίου Πειραιώς δεν υποδηλώνει αποδοχή των γνώμων του συγγραφέα.

*Στους γονείς μου,
Μανώλη και Ευγενία.
Στη γιαγιά μου,
Πόπη.*

Ευχαριστίες

Εκφράζω από καρδιάς τις ευχαριστίες μου στον επιβλέποντα καθηγητή μου κ. Πελέκη Νικόλαο ο οποίος προσέφερε πολύτιμη και ουσιαστική καθοδήγηση κατά τη διαδικασία εκπόνησης της εν λόγω εργασίας. Ακόμη, θέλω να ευχαριστήσω και το υπόλοιπο διδακτικό προσωπικό του μεταπτυχιακού μου προγράμματος, που συνετέλεσε στη θεμελίωση των γνώσεων που απαιτούνται για την απόκτηση του μεταπτυχιακού διπλώματος.

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Περίληψη

Η παρούσα διπλωματική εργασία εξετάζει την πρόβλεψη τροχιάς αεροσκαφών σε τρισδιάστατο χώρο με χρήση δεδομένων Automatic Dependent Surveillance–Broadcast (ADS-B) και μοντέλων Long Short-Term Memory (LSTM). Στόχος είναι η ανάπτυξη και αξιολόγηση ενός δεδομενοκεντρικού πλαισίου για πρόβλεψη της μελλοντικής μετατόπισης αεροσκαφών σε χρονικούς ορίζοντες 5, 10 και 15 λεπτών.

Τα δεδομένα προέρχονται από το *OpenSky Network* και υποβλήθηκαν σε καθαρισμό, φιλτράρισμα, χρονικό διαχωρισμό και κατασκευή ακολουθιών πρόβλεψης. Εξετάστηκαν τόσο βασικά χαρακτηριστικά θέσης όσο και εμπλουτισμένη αναπαράσταση με τη συμπερίληψη ταχύτητας, κατακόρυφου ρυθμού και ημιτονοειδούς/συνημιτονοειδούς κωδικοποίησης της διεύθυνσης κίνησης. Η απόδοση αξιολογήθηκε με οριζόντια και κατακόρυφα σφάλματα, χρησιμοποιώντας μέσο, διάμεσο και 90° (άνω) εκατοστημόριο (p90).

Τα αποτελέσματα έδειξαν ότι ο εμπλουτισμός των χαρακτηριστικών μειώνει το μέσο και το p90 οριζόντιο σφάλμα σε όλους τους εξεταζόμενους χρονικούς ορίζοντες. Η σύγκριση διαφορετικών LSTM αρχιτεκτονικών οδήγησε στην επιλογή του ENRICH_LSTM96 ως της πιο ισορροπημένης διαμόρφωσης για την τελική αξιολόγηση. Το επιλεγμένο μοντέλο εξετάστηκε επιπλέον σε χρονικά διακριτό πλήρες ημερήσιο σύνολο δεδομένων και παρουσίασε σχετικά σταθερή απόδοση κατά τη διάρκεια της ημέρας.

Τέλος, από τις προβλεπόμενες μετατοπίσεις υπολογίστηκε ένας παράγωγος χρονικός δείκτης τύπου ETA, με σκοπό τη διερεύνηση της σχέσης μεταξύ χωρικού και χρονικού σφάλματος. Ο δείκτης αυτός δεν αποτελεί συμβατική πρόβλεψη χρόνου άφιξης σε συγκεκριμένο προορισμό, αλλά παρέχει συμπληρωματική χρονική ερμηνεία των αποτελεσμάτων πρόβλεψης τροχιάς.

Επιστημονική Περιοχή: Μηχανική Μάθηση, Ανάλυση Δεδομένων, Αεροπορικά Συστήματα

Λέξεις Κλειδιά: Trajectory Prediction, ADS-B, LSTM, Air Traffic Management, ETA-style Temporal Metric

Abstract

This thesis investigates aircraft trajectory prediction in three-dimensional space using Automatic Dependent Surveillance–Broadcast (ADS-B) data and Long Short-Term Memory (LSTM) models. The objective is to develop and evaluate a data-driven framework for predicting future aircraft displacement at 5, 10, and 15-minute horizons.

ADS-B data from the *OpenSky Network* were processed through cleaning, filtering, temporal segmentation, and sequence construction. Both positional features and an enriched representation including ground velocity, vertical rate, and sine-cosine heading encoding were examined. Model performance was evaluated using horizontal and vertical errors summarized by mean, median, and 90th-percentile (p90) measures.

The results showed that feature enrichment reduced mean and p90 horizontal error across all examined horizons. A comparison of alternative LSTM architectures led to the selection of ENRICH_LSTM96 as the most balanced configuration for final evaluation. The selected model was additionally tested on a temporally distinct full-day ADS-B dataset, where its hourly performance remained broadly stable.

Finally, an ETA-style temporal metric was derived from the predicted displacement to examine the relationship between spatial and temporal prediction error. This quantity is not a conventional destination-specific Estimated Time of Arrival prediction, but rather a supplementary temporal interpretation of the trajectory prediction results.

Scientific Area: Machine Learning, Data Analysis, Air Traffic Systems

Key Words: Trajectory Prediction, ADS-B, LSTM, Air Traffic Management, ETA-style Temporal Metric

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Table of Abbreviations – Acronyms

ADS-B	Automatic Dependent Surveillance–Broadcast
ATM	Air Traffic Management
TBO	Trajectory-Based Operations
ETA	Estimated Time of Arrival
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
ML	Machine Learning
AI	Artificial Intelligence
GPS	Global Positioning System
ICAO	International Civil Aviation Organization
FL	Flight Level
CSV	Comma-Separated Values
p90	90th Percentile
km	Kilometer
m	Meter
min	Minute

Introduction

Air Traffic Management (ATM) systems increasingly rely on accurate information about the expected future motion of aircraft. Within the broader transition toward Trajectory-Based Operations (TBO), trajectory prediction supports applications such as conflict detection, traffic-flow management, arrival sequencing, separation assurance, and operational decision support. The ability to estimate future aircraft position over short and medium forecasting horizons is therefore an important component of modern air traffic analysis and planning.

The increasing availability of Automatic Dependent Surveillance-Broadcast (ADS-B) data has created new opportunities for data-driven trajectory prediction. ADS-B observations provide high-frequency information about aircraft position and motion, including latitude, longitude, altitude, velocity, heading, and vertical rate. Large-scale surveillance repositories such as the *OpenSky Network* make it possible to investigate aircraft behavior using real operational trajectories rather than exclusively relying on simulated or performance-model-based data.

Aircraft trajectory prediction nevertheless remains challenging. Future motion is influenced by factors such as route intent, Air Traffic Control interventions, weather, aircraft performance, and surrounding traffic, many of which may not be directly available from surveillance data alone. Prediction uncertainty also increases with the forecasting horizon, as the aircraft has more opportunity to change speed, heading, altitude, or route.

Machine learning approaches offer a way to learn relationships between historical surveillance states and future aircraft motion directly from observed trajectories. In particular, recurrent neural network architectures such as Long Short-Term Memory (LSTM) networks are suitable for sequential data because they can represent temporal dependencies across successive observations. The present study investigates this approach in a deliberately constrained setting based only on ADS-B state-vector information and derived kinematic features. A detailed review of classical, data-driven, recurrent, and context-aware trajectory prediction methods is provided in Chapter 2.

The thesis focuses on deterministic three-dimensional endpoint prediction at predefined future horizons. Historical aircraft observations are used to estimate future displacement in latitude, longitude, and altitude at 5, 10, and 15-minute elapsed time

horizons. Particular attention is given to the effect of kinematic feature enrichment, alternative compact LSTM architectures, horizon-dependent prediction performance, and evaluation on a temporally distinct full-day dataset.

Objectives of the study

The primary objective of this thesis is to develop and evaluate a data-driven framework for aircraft trajectory prediction in three-dimensional space using ADS-B state-vector data. More specifically, the study aims to construct model-ready trajectory datasets through data cleaning, temporal segmentation, feature engineering, and timestamp-based future target construction. It evaluates LSTM-based models for predicting future aircraft displacement and investigates whether enriching positional information with velocity, heading, and vertical-rate variables improves predictive performance.

A further objective is to compare alternative compact LSTM configurations in order to examine the effect of recurrent capacity and depth under a common experimental framework. Model performance is evaluated at 5, 10, and 15-minute horizons using horizontal and vertical error measures, including upper-percentile statistics designed to characterize less favorable prediction cases.

Finally, the selected model is evaluated on a temporally distinct full-day ADS-B dataset to assess the stability of prediction performance beyond the initial development data. The study additionally derives an ETA-style temporal metric from the predicted motion in order to examine the relationship between spatial and temporal prediction error.

Contributions of the Thesis

The contribution of this thesis lies in the structured empirical evaluation of an ADS-B state-vector-based trajectory prediction framework under clearly defined methodological constraints. The study develops a complete processing pipeline from raw surveillance observations to horizon-specific supervised prediction datasets and evaluates both positional and kinematically enriched input representations.

A comparative analysis of compact LSTM architectures is conducted in order to examine the effects of recurrent capacity, architectural depth, and dropout regularization under a common experimental setup. Prediction performance is

assessed using physically interpretable horizontal and vertical error metrics across multiple elapsed-time horizons.

An additional contribution is the evaluation of the selected horizon-specific models on a temporally distinct full-day ADS-B dataset. This experiment provides a robustness assessment beyond the limited development data and allows prediction stability to be examined across hourly variations in observed traffic.

Finally, the thesis investigates a derived ETA-style temporal metric based on the predicted displacement. This metric provides an additional interpretation of the relationship between spatial and temporal prediction error, while remaining distinct from conventional destination-specific ETA prediction.

Structure of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 1** defines the trajectory prediction problem, the modeling rationale, the evaluation perspective, and the scope of the study.
- **Chapter 2** presents the literature review and methodological background, covering aircraft trajectory prediction, ADS-B surveillance, classical and data-driven approaches, recurrent neural networks, trajectory representation, evaluation practices, and context-aware prediction.
- **Chapter 3** describes the ADS-B data source, preprocessing pipeline, feature engineering, sliding-window construction, and statistical characteristics of the experimental dataset.
- **Chapter 4** presents the LSTM architecture family, training strategy, regularization, multi-horizon modeling, and model selection procedure.
- **Chapter 5** describes the experimental design, training protocol, reconstruction of predicted positions, and evaluation metrics.
- **Chapter 6** presents the experimental results, including baseline performance, feature enrichment, architecture comparison, full-day robustness evaluation, and derived ETA analysis.
- **Chapter 7** discusses the principal findings, limitations, future research directions, and conclusions of the study.

Chapter 1: Analytical Description of the subject

1.1 Background and context

Aircraft trajectory prediction is an important component of modern Air Traffic Management, supporting the estimation of future aircraft motion for operational monitoring and decision-support applications. The present thesis focuses specifically on data-driven prediction of aircraft movement using Automatic Dependent Surveillance–Broadcast (ADS-B) state-vector observations.

The study considers aircraft operating above Flight Level 200 within the defined European geographical region and investigates prediction at elapsed-time horizons of 5, 10, and 15 minutes. Rather than incorporating flight plans, meteorological information, aircraft-performance models, or explicit Air Traffic Control intent, the framework relies exclusively on historical surveillance states and derived kinematic features.

This restricted information setting is intentional. It allows the predictive value of ADS-B state histories to be examined directly and provides a controlled environment for evaluating the effects of feature representation, recurrent model capacity, prediction horizon, and temporal robustness.

The prediction problem is formulated as deterministic three-dimensional endpoint forecasting. Historical aircraft observations are supplied to Long Short-Term Memory (LSTM) sequence models, which estimate the future displacement of the aircraft in latitude, longitude, and altitude. The complete methodological and literature background supporting this formulation is presented in Chapter 2.

1.2 Formal Problem Definition

An aircraft trajectory may be defined as a temporarily ordered sequence of state vectors:

$$T = \{s_1, s_2, \dots, s_T\} \quad (1.1)$$

where each state vector is given by:

$$s_t = (lat_t, lon_t, alt_t, u_t, ur_t, y_t) \quad (1.2)$$

of which:

- lat_t, lon_t denote geographic coordinates,
- alt_t denotes altitude,
- u_t denotes ground speed magnitude,
- ur_t denotes vertical rate,
- y_t denotes heading angle

For model input, the observed states are transformed into the enriched feature representation

$$z_t = (\Delta lat_t, \Delta lon_t, \Delta alt_t, u_t, \sin y_t, \cos y_t, ur_t) \quad (1.3)$$

where the first three terms represent first-order positional differences.

Given a fixed length historical window of L observations:

$$X_t = \{z_{t-L+1}, \dots, z_t\} \quad (1.4)$$

the objective is to predict aircraft displacement after a fixed horizon of H seconds:

$$\Delta p_{t+H} = (\Delta lat, \Delta lon, \Delta alt)_{t+H} \quad (1.5)$$

The prediction task is therefore formulated as a supervised regression problem:

$$f_\theta(X_t) = \widehat{\Delta p}_{t+H} \quad (1.6)$$

where:

- f_θ is a parameterized neural network,
- θ denotes model parameters,
- $\widehat{\Delta p}_{t+H}$ is a predicted displacement vector.
- The model is trained to minimize a loss function over N training samples:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N \left\| \Delta p_{t+H}^{(i)} - \widehat{\Delta p}_{t+H} \right\|^2 \quad (1.7)$$

which corresponds to the Mean Square Error (MSE) between true and predicted displacements.

This formulation treats trajectory prediction as a sequence to one regression problem. Separate models are trained and evaluated for each prediction horizon $H \in \{300, 600, 900\}$ seconds. Because ADS-B observations are not sampled at perfectly uniform temporal intervals, the future target state was selected using the observation timestamps rather than a fixed row-index offset. For each current state at time t , the target was defined as the first available observation whose timestamp corresponded to $t + H$, subject to a maximum temporal tolerance of 10 seconds. Samples for which no suitable future observation was available were excluded.

In addition to predicting the horizontal displacement of the aircraft, the prediction task also includes the estimation of vertical motion through the altitude component of the displacement vector. As a result, the proposed framework allows the evaluation of trajectory prediction accuracy in both horizontal and vertical dimensions. This dual perspective is particularly important for enroute operations where aircraft motion evolves simultaneously across spatial and vertical axes, and both components contribute to the overall trajectory evolution.

1.3 Data-Driven Modeling Rationale

The present study adopts a data-driven modeling approach in order to predict future aircraft motion directly from historical ADS-B state-vector observations. This choice is consistent with the objective of evaluating trajectory prediction in a restricted information setting, without relying on flight plans, meteorological data, aircraft-performance tables, or explicit Air Traffic Control intent.

Long Short-Term Memory (LSTM) networks are used because the prediction problem is sequential: the future displacement of an aircraft depends on the evolution of its recent motion rather than on a single instantaneous observation. The theoretical motivation for recurrent and LSTM-based trajectory modeling is discussed in detail in Chapter 2.

To represent aircraft motion, the study uses a delta-based formulation in which positional changes between consecutive observations are used as model inputs and future displacement is used as the prediction target. This representation emphasizes

local motion dynamics rather than absolute geographic position and is combined, in the enriched models, with ground velocity, vertical rate, and sine-cosine heading features.

The resulting formulation provides a controlled framework for examining the effect of feature enrichment, recurrent architecture, and prediction horizon on three-dimensional aircraft trajectory prediction.

1.4 Operational Evaluation Framework

A distinctive aspect of this study is the evaluation of robustness under full day operational variability. Rather than reporting performance on isolated trajectory segments, the selected model is evaluated across 24 consecutive hourly files corresponding to May 2, 2019. This full-day evaluation allows the stability of prediction performance to be examined across hourly variations in the observed traffic and beyond the limited development data.

Additionally, a derived ETA-style temporal metric is computed from the predicted motion. Although arrival time is not directly predicted by the model, this quantity provides a secondary temporal interpretation of the trajectory prediction output and allows the relationship between spatial and temporal prediction error to be examined.

1.5 Scope and Boundaries

The scope of this thesis is explicitly constrained as follows:

- Only ADS-B state vector data are used.
- No flight plan, meteorological, or aircraft performance tables are incorporated.
- No hybrid physics-ML baseline is implemented.
- Only LSTM architectures are evaluated.
- Prediction is limited to fixed horizons of up to 15 minutes.

The study does not attempt long-horizon and full-flight trajectory reconstruction, intent inference or real time deployment optimization.

Chapter 2: Literature Review and Methodological Background

2.1 Aircraft Trajectory Prediction in Air Traffic Management and Trajectory-Based Operations

Aircraft trajectory prediction is a fundamental component of modern Air Traffic Management (ATM), as many operational functions depend on an estimate of how an aircraft state is expected to evolve over time. Predicted trajectories support applications such as conflict detection, traffic-flow management, sequencing, separation assurance, arrival management, and the broader transition toward Trajectory-Based Operations (TBO). The required prediction horizon, level of detail, and accuracy depend strongly on the operational task for which the trajectory is generated (EUROCONTROL, 2017).

Aircraft trajectory prediction does not correspond to a single uniformly defined forecasting problem. Different studies address different spatial and temporal scales, ranging from short-horizon tactical prediction to longer-term planning and arrival management applications. Similarly, some methods estimate a single future aircraft state, whereas others generate a complete sequence of future trajectory points. Three-dimensional prediction typically concerns horizontal position and altitude, while four-dimensional formulations additionally incorporate time explicitly. These differences affect both model design and the interpretation of prediction performance.

Trajectory prediction is also influenced by uncertainty in future aircraft behavior. Current motion alone may not fully determine the future path, since aircraft trajectories are affected by pilot actions, ATC instructions, aircraft performance, weather, route constraints, surrounding traffic, and operational procedures. As the prediction horizon increases, uncertainty generally grows because a larger number of future deviations may occur.

Most traditional trajectory predictors produce a deterministic estimate of the future aircraft state. ADS-B-based trajectory prediction has, for example, been studied in the context of conflict detection, demonstrating the direct relevance of future-state estimation to ATM decision support (Baek & Bang, 2012). Probabilistic approaches extend this concept by representing a distribution of plausible future trajectories rather than a single point estimate. Barratt, Kochenderfer, and Boyd (2019), for instance,

developed probabilistic aircraft trajectory models from historical terminal-airspace position data, illustrating how uncertainty in future motion can be represented explicitly.

The aircraft trajectory-prediction literature therefore spans a broad range of approaches, including kinematic extrapolation, aircraft-performance models, probabilistic methods, data-driven learning, recurrent neural networks, and more recent context-aware architectures. These methods differ substantially in their information requirements, assumptions, computational complexity, and intended operational use. Consequently, direct numerical comparison between studies is often difficult because datasets, flight phases, prediction horizons, preprocessing procedures, input variables, and evaluation metrics vary considerably.

Within this broader landscape, the present thesis focuses on deterministic, data-driven prediction of future three-dimensional aircraft position from ADS-B state-vector histories. Rather than reconstructing an entire future flight path, the study adopts a sequence-to-one formulation in which a historical observation sequence is used to estimate future aircraft displacement at predefined elapsed-time horizons of 5, 10, and 15 minutes. These horizons should be interpreted as experimental settings of the present study rather than as universal definitions of short or medium-term trajectory prediction.

The study intentionally operates in a comparatively low-information setting. Prediction is based on surveillance state histories and derived kinematic features, without incorporating flight plans, weather, aircraft-performance tables, or explicit ATC intent. The following sections therefore review the progression from ADS-B surveillance and classical trajectory-prediction approaches to data-driven and recurrent neural network methods, before examining the representation, training, evaluation, and contextual factors most directly related to the methodology adopted in this thesis.

2.2 ADS-B Surveillance and the OpenSky Network

Automatic Dependent Surveillance–Broadcast (ADS-B) has become an important source of aircraft surveillance information for operational monitoring and aviation research. Aircraft equipped with ADS-B periodically broadcast state information derived from onboard navigation and avionics systems, including identification, position, altitude, ground speed, heading, and vertical motion. These observations provide a

detailed temporal record of realized aircraft behavior and are therefore well suited to data-driven trajectory analysis.

A major research infrastructure for collecting ADS-B data is the *OpenSky Network*. Schäfer et al. (2014) introduced OpenSky as a large-scale, crowdsourced surveillance network that aggregates aircraft messages from geographically distributed receivers. The platform has since become an important source of operational air traffic data for research in trajectory reconstruction, anomaly detection, aircraft-performance analysis, and trajectory prediction.

The use of crowdsourced surveillance data offers substantial advantages but also introduces methodological challenges. Strohmeier et al. (2021) documented that OpenSky datasets are affected by receiver coverage, message availability, aircraft equipment, timestamp handling, and the construction of derived datasets. Consequently, ADS-B observations should not automatically be treated as perfectly uniform time series. Temporal gaps, duplicated observations, missing fields, and variations in sampling density may occur and can directly affect the construction of prediction sequences.

These characteristics make preprocessing an integral part of ADS-B-based trajectory prediction. Before model-ready samples can be generated, aircraft observations must be ordered temporally, associated with the correct aircraft, checked for required variables, and divided into sufficiently continuous trajectory segments. Different studies address irregular or incomplete trajectories in different ways. Wu et al. (2022), for example, combine cleaning, segmentation, interpolation, normalization, clustering, and sliding-window construction within an OpenSky-based prediction workflow. Lin et al. (2021) address missing trajectory observations explicitly through trajectory imputation. These approaches illustrate that surveillance-data preprocessing is a methodological choice rather than a purely technical one.

The OpenSky Network also provides broad spatial and temporal coverage, but the resulting datasets remain dependent on the distribution and availability of receivers. For this reason, ADS-B prediction studies should clearly define their geographical region, altitude range, time period, flight phase, and filtering criteria. These decisions influence the composition of the resulting trajectory dataset and therefore the interpretation of predictive performance.

The present thesis follows a direct surveillance-driven approach. OpenSky state-vector observations are cleaned and segmented into temporally continuous aircraft trajectories, while only observations containing the variables required for the prediction model are retained. Rather than reconstructing all trajectories onto a uniform temporal grid, the study preserves the observed sequence structure and defines future targets using elapsed time. The detailed implications of irregular sampling, trajectory segmentation, sliding-window construction, and feature representation are discussed later in Section 2.6.

2.3 Classical, Model-Based and Hybrid Trajectory Prediction

Before the widespread adoption of machine learning methods, aircraft trajectory prediction was primarily addressed through kinematic, state-estimation, and aircraft performance-based approaches. These methods propagate the current aircraft state into the future using explicit assumptions about motion, dynamics, performance, and, where available, operational intent. Their main advantage is interpretability, but their predictive accuracy depends strongly on the quality of the information supplied to the model.

The simplest class of predictors is based on kinematic extrapolation. Current or estimated quantities such as speed, heading, vertical rate, and acceleration are propagated forward under assumptions of sufficiently stable motion. Such models are computationally efficient and can be useful over short tactical horizons, but their accuracy deteriorates when aircraft behavior deviates from the assumed motion pattern through turns, climbs, descents, or ATC interventions.

State-estimation methods extend this principle by combining noisy surveillance observations with one or more candidate motion models. Baek and Bang (2012), for example, investigated ADS-B-based trajectory prediction for conflict-detection applications, illustrating how surveillance-derived future-state estimates can directly support ATM decision making.

A more detailed class of predictors uses aircraft-performance and physics-based models. These methods may incorporate point-mass or energy balance equations together with aircraft performance information, atmospheric conditions, and operational intent. Frameworks based on aircraft-performance databases such as BADA can produce physically meaningful trajectories, but important parameters required by

ground-based predictors may be unavailable or uncertain. Aircraft mass, thrust settings, speed intent, future clearances, and meteorological conditions can all influence the resulting trajectory.

Alligier, Gianazza, and Durand (2013) demonstrated how data-driven estimation can be combined with such physical models. Their approach used historical observations to estimate unavailable aircraft mass and thrust-related information in order to improve climb-trajectory prediction. Rather than replacing the underlying aircraft-performance model, machine learning was used to estimate uncertain operational quantities required by it.

Alligier and Gianazza (2018) extended this idea in a large-scale OpenSky-based study covering multiple airports and aircraft types. Learned operational factors were incorporated into a performance-based climb predictor, producing substantial improvements compared with baseline assumptions. This work illustrates the potential of hybrid physics-data approaches, which combine the interpretability of structured aircraft models with empirical estimation of uncertain parameters.

These model families differ mainly in the information they require. Kinematic predictors need relatively little information but become unreliable when motion changes. Performance-based methods can incorporate aircraft capabilities and operational constraints but require accurate model parameters and contextual inputs. Hybrid approaches reduce some of these limitations by estimating unavailable quantities from surveillance data, at the cost of additional modeling complexity.

This trade-off provides an important motivation for purely data-driven trajectory prediction. When large surveillance datasets are available but detailed aircraft-performance, weather, flight-plan, or intent information is not, a predictive model can instead attempt to learn future motion directly from historical state sequences. Such models sacrifice part of the physical interpretability of model-based approaches but reduce dependence on information that may not be accessible in a surveillance-only setting.

The present thesis adopts this lower-information formulation. Prediction relies on ADS-B state-vector histories and derived kinematic features, without using BADA performance tables, aircraft mass, weather forecasts, flight plans, or explicit ATC intent. The objective is therefore not to reproduce a fully informed operational trajectory

predictor, but to evaluate how much predictive information can be extracted from recent surveillance states alone.

The next section examines how this shift toward empirical modeling developed from trajectory clustering and classical machine-learning methods into modern deep sequence-learning approaches.

2.4 From Trajectory Mining to Machine and Deep Learning

The increasing availability of large surveillance datasets encouraged a gradual shift from explicit model-based trajectory prediction toward empirical methods that learn recurring patterns directly from historical aircraft movement. Rather than relying exclusively on aircraft-performance equations or detailed operational parameters, data-driven approaches attempt to extract statistical regularities from previously observed trajectories.

An important early direction was trajectory mining and clustering. Historical aircraft tracks often exhibit recurring structures because operations are constrained by routes, procedures, airspace organization, and airport configurations. Gariel, Srivastava, and Feron (2011) demonstrated that recorded trajectories can be grouped into representative patterns useful for characterizing nominal airspace behavior and identifying deviations. Although clustering does not by itself constitute fixed horizon future-state prediction, it established the broader principle that surveillance trajectories contain exploitable empirical structure.

Classical machine learning methods subsequently extended this idea by learning mappings between observed aircraft states and future quantities. Regression, probabilistic models, Hidden Markov Models, tree-based methods, and support-vector approaches offered greater flexibility than purely kinematic extrapolation while avoiding the full information requirements of aircraft-performance models. Their main limitation, however, was often the need for substantial manual feature engineering and explicit representation of temporal history.

This limitation motivated the adoption of models designed specifically for sequential data. Feed-forward neural networks can approximate nonlinear mappings but do not inherently retain temporal memory. Recurrent Neural Networks (RNNs), in contrast,

process ordered observations through an evolving hidden state, making them more naturally suited to trajectory histories.

Probabilistic data-driven approaches also emerged as an important complementary direction. Barratt, Kochenderfer, and Boyd (2019), for example, learned probabilistic trajectory models from historical aircraft position data in terminal airspace. Their work demonstrates that surveillance data can be used not only to estimate a single future trajectory but also to represent uncertainty over multiple plausible future motions.

More recent trajectory-prediction studies increasingly combine deep sequence learning with spatial, geometric, contextual, or attention-based mechanisms. CNN-LSTM architectures, bidirectional recurrent networks, GRU-based models, attention mechanisms, and Transformer-style approaches reflect the growing flexibility of the field. However, these methods should not be interpreted as universally superior to simpler models, since reported performance depends strongly on the available inputs, prediction horizon, preprocessing procedure, operational context, and evaluation protocol.

A further development is the growing recognition that model architecture and trajectory representation should be considered jointly. Modern systems frequently combine learned temporal representations with explicit kinematic, geometric, or contextual features. Consequently, the distinction between handcrafted feature engineering and representation learning is increasingly blurred in practical aircraft trajectory-prediction systems.

Within this methodological evolution, the present thesis adopts a compact recurrent formulation. Historical ADS-B state vectors and derived kinematic features are supplied directly to LSTM-based sequence models that estimate future three-dimensional displacement at predefined elapsed-time horizons. The study therefore belongs to the deep sequence-learning branch of the literature, while intentionally avoiding richer inputs such as flight plans, weather, route semantics, or multi-aircraft interaction models.

The next section focuses specifically on Recurrent Neural Networks and Long Short-Term Memory architectures, which form the principal modeling framework of the present study.

2.5 Recurrent Neural Networks and LSTM for Aircraft Trajectories

Aircraft trajectory data are inherently sequential, since the state of an aircraft at a given instant depends strongly on its preceding motion. Variables such as position, speed, heading, and vertical rate evolve over time, making trajectory prediction different from conventional regression problems in which observations can be treated as independent feature vectors. A suitable model must therefore incorporate information from a sequence of past observations rather than relying only on the current state.

Recurrent Neural Networks (RNNs) are designed for this purpose. Unlike feed-forward networks, an RNN maintains a hidden state that is updated as successive observations are processed. This recurrent state allows information from earlier time steps to influence later predictions and provides a natural framework for representing temporal dependencies in aircraft motion.

A major limitation of conventional RNNs is the difficulty of learning long-range dependencies. During backpropagation through time, gradients may become progressively smaller or larger, leading to the well-known vanishing and exploding-gradient problems. Bengio, Simard, and Frasconi (1994) demonstrated the difficulty of learning long-term dependencies with standard gradient-based recurrent architectures, motivating the development of improved recurrent mechanisms.

Long Short-Term Memory (LSTM) networks were introduced by Hochreiter and Schmidhuber (1997) to address these limitations. LSTM incorporates an internal memory state and gating mechanisms that regulate information flow through the network. In modern formulations, the input gate controls how much new information is stored, the forget gate regulates how much previous information is retained, and the output gate determines how much of the memory state contributes to the hidden representation.

These mechanisms make LSTM particularly suitable for trajectory prediction because aircraft motion often exhibits temporal persistence. Heading, velocity, and vertical rate may remain relatively stable over a sequence of observations, while changes in these variables can indicate the onset of a maneuver. By processing an ordered history of states, an LSTM can learn both short-term motion changes and longer temporal patterns that are difficult to represent with a single instantaneous observation.

The application of LSTM to aircraft trajectory prediction has been demonstrated directly in the aviation literature. Shi et al. (2018) used ADS-B trajectory data and sliding-window sequence construction in an LSTM-based flight trajectory-prediction framework. Their study considered three and four-dimensional prediction settings and reported improved performance relative to Markov-based comparison methods. This work provides an important aviation-specific precedent for using recurrent sequence models to learn future motion from historical surveillance trajectories.

Subsequent studies introduced more complex recurrent and hybrid architectures. Ma and Tian (2020), for example, combined convolutional and LSTM components for aircraft four-dimensional trajectory prediction, using convolutional layers to extract spatial characteristics and LSTM to model temporal dependencies. Such approaches illustrate the broader evolution of the field toward architectures that combine temporal sequence learning with additional spatial or contextual representations.

An important architectural choice concerns the number and size of recurrent layers. A single-layer LSTM provides a compact temporal encoder, while a stacked LSTM processes the learned sequence representation through multiple recurrent layers. Greater depth or a larger hidden-state dimension can increase representational capacity, but also introduces additional parameters, computational cost, and potential overfitting. Consequently, recurrent depth and hidden-state width are typically treated as empirical design choices rather than universally optimal settings.

The output of the recurrent encoder is often passed through one or more fully connected layers before the final prediction. These dense layers act as a regression head that maps the learned temporal representation to the required target variables. In aircraft trajectory prediction, this architecture allows the recurrent layers to encode motion history while the final dense layers transform that representation into future spatial coordinates or displacements.

Trajectory-prediction models can also differ in the structure of their outputs. In sequence-to-one forecasting, a complete historical sequence is mapped to a single future state or displacement vector. In sequence-to-sequence forecasting, the model predicts a series of future states representing an entire future trajectory segment. The appropriate formulation depends on the research objective: sequence-to-one models are suitable for evaluation at specific future horizons, whereas sequence-to-sequence models are more appropriate when the complete future path is required.

Regularization is also important when training recurrent neural networks. Dropout, introduced by Srivastava et al. (2014), reduces overfitting by randomly deactivating a fraction of neural activations during training. It can be applied between recurrent or dense layers and is commonly used to improve model robustness. The general principle of dropout is well established, but the exact dropout probability remains a task-specific hyperparameter rather than a theoretically prescribed value.

Alternative gated recurrent architectures, such as the Gated Recurrent Unit (GRU), provide similar temporal-memory functionality with a simpler internal structure. More recent trajectory-prediction systems also combine recurrent networks with attention mechanisms, temporal convolutions, or Transformer-based components. These alternatives demonstrate that no single recurrent architecture is universally optimal; performance depends on the dataset, representation, prediction horizon, and available contextual information.

Recent research has further extended recurrent prediction beyond the isolated history of a single aircraft. Social-LSTM approaches incorporate information from neighboring aircraft, while semantic and flight-plan-aware models use route or intent information. These methods can resolve ambiguities that may not be inferable from a state-vector history alone, but they also require richer input data and more complex modeling frameworks.

Within this literature, the present thesis adopts a deliberately compact LSTM-based design. Historical ADS-B observations are processed as ordered sequences, and the recurrent representation is used to predict a three-dimensional future displacement. Both single-layer and stacked LSTM configurations are evaluated in order to examine the effect of recurrent capacity and depth under a common input representation and training procedure.

The evaluated architectures should be interpreted as study-specific experimental configurations, not as direct reproductions of a single published model. The literature supports the general use of LSTM sequence modeling, recurrent depth, variation in hidden-state capacity, dense regression heads, and dropout regularization, but it does not establish the exact 64-unit, 96-unit, or 64→32 configurations used in this thesis as canonical architectures.

The architectural comparison is therefore empirical in nature. By altering recurrent capacity and depth while keeping the broader experimental setup consistent, the study

examines whether increased model complexity provides measurable benefits for the available ADS-B prediction task.

The effectiveness of the LSTM framework, however, also depends strongly on how the trajectory is represented before it reaches the network. Coordinate formulation, kinematic feature enrichment, temporal segmentation, window construction, scaling, and target definition can all influence the learning problem. These aspects are examined in the following section.

2.6 Trajectory Representation, Feature Engineering and Sequence Construction

The performance of an aircraft trajectory-prediction model depends not only on the learning algorithm but also on how the trajectory is represented. Raw surveillance observations must be transformed into structured inputs and prediction targets before they can be used by a machine learning model. Coordinate formulation, derived kinematic variables, temporal continuity, sequence construction, sampling irregularity, and scaling therefore constitute important methodological choices.

A fundamental distinction concerns the use of absolute positions versus relative motion representations. Absolute coordinates preserve the geographical location of the aircraft but require the model to learn both spatial context and local motion dynamics. Relative representations instead describe changes with respect to a previous state or reference point and can expose the local structure of motion more directly.

Recent trajectory-prediction research increasingly treats coordinate representation as part of the modeling problem itself. Geometry-aware approaches, including recent ADS-B studies, use local or relative coordinate systems together with kinematic and geometric quantities so that displacement, direction, and local motion relationships are represented more explicitly. This perspective provides methodological support for displacement-based prediction, in which the future change in aircraft state is modeled rather than the complete absolute position directly.

The present thesis follows this general principle by predicting a three-dimensional displacement in latitude, longitude, and altitude. The exact target definition is specific to the experimental design of this study rather than a standardized representation prescribed by previous literature. The broader motivation, however, is consistent with

the idea that relative representations can simplify the learning of local trajectory dynamics.

Kinematic feature enrichment

Aircraft position alone does not fully describe motion. Variables such as velocity, heading, and vertical rate provide direct information about the direction and magnitude of aircraft movement and are therefore frequently incorporated into trajectory-prediction models.

Wu et al. (2022), for example, use a preprocessing and representation pipeline that combines trajectory reconstruction, normalization, clustering, and spatiotemporal feature modeling before neural prediction. More recent representation-aware approaches likewise combine learned sequence representations with explicit geometric or kinematic variables.

The present study enriches the positional representation with ground velocity, vertical rate, and heading information. Heading is encoded through its sine and cosine components because it is a circular variable: values near 0° and 360° represent nearly identical directions despite being numerically far apart. The sine-cosine transformation removes this discontinuity and provides a continuous representation suitable for gradient-based learning.

The exact combination of features used in this thesis should be interpreted as a study-specific design choice. The literature supports the broader value of combining positional and kinematic information, but it does not prescribe a single universally optimal aircraft trajectory feature set.

Temporal continuity and irregular sampling

Sequential prediction requires observations to correspond to temporally meaningful trajectory segments. ADS-B records may contain interruptions, coverage gaps, or irregular temporal spacing, meaning that adjacent records do not always correspond to a continuous motion sequence.

Trajectory segmentation is therefore commonly used to prevent observations separated by large temporal gaps from being treated as consecutive points of one

continuous trajectory. Wu et al. (2022), among others, incorporate segmentation and trajectory reconstruction before supervised samples are generated. This highlights the importance of defining temporal continuity before recurrent modeling is applied.

Irregular sampling also affects the interpretation of sequence length and prediction horizon. A fixed number of consecutive observations does not necessarily correspond to a fixed duration unless the sampling interval is perfectly uniform. Some studies address this issue through interpolation or regular resampling, thereby reconstructing trajectories on an approximately uniform temporal grid.

The present thesis instead preserves consecutive observed states and defines the future prediction target according to elapsed time rather than row position. This distinction is important because it separates the amount of historical information presented to the model from the actual temporal duration of the forecast horizon.

Sliding-window sequence construction

Sliding windows are widely used to transform continuous trajectories into supervised learning samples. A historical sequence of observations is used as model input, while one or more future states provide the corresponding target. Shi et al. (2018), for example, employed sliding-window sequence construction in LSTM-based aircraft trajectory prediction.

This approach preserves temporal ordering and allows long trajectories to produce multiple training samples. However, neighboring windows may overlap substantially and therefore share many of the same observations. Consequently, dataset construction and train-test partitioning must be interpreted carefully, particularly when random window-level splits are used.

The present study employs window-based sequence construction during model development and complements the internal evaluation with a temporally distinct later-day dataset. The implications of this evaluation design are discussed further in Section 2.7.

Scaling and numerical representation

Trajectory features and prediction targets may differ considerably in numerical scale and physical units. Without normalization, variables with larger numerical magnitudes may exert disproportionate influence on optimization.

Standardization is therefore commonly applied before neural network training. Preprocessing transformations should be estimated using the training data only and then applied unchanged to validation and test datasets in order to avoid information leakage.

This principle is particularly relevant for multi-output displacement prediction, where horizontal coordinate changes and altitude differences are represented on different numerical scales. Predictions can subsequently be transformed back to their original units before physical-domain evaluation.

Relationship to the present thesis

The representation strategy used in this thesis combines several principles supported by previous trajectory prediction research: relative displacement modeling, kinematic feature enrichment, segmentation of temporally continuous observations, sliding-window sequence construction, explicit handling of irregular sampling, and standardized numerical inputs.

At the same time, the exact implementation remains study-specific. The selected historical sequence length, feature combination, three-dimensional displacement target, and timestamp-based future-state selection were defined for the experimental framework of the present work rather than reproduced directly from a single previous study.

The following section examines the training and evaluation practices used to assess such sequence models, with particular emphasis on optimization, regularization, prediction horizons, error metrics, and generalization.

2.7 Training, Evaluation and Generalization Practices

The performance of an aircraft trajectory-prediction model depends not only on its architecture and input representation but also on the training and evaluation framework. Optimization strategy, loss function, regularization, prediction horizon, data partitioning, and performance metrics all affect how model results should be interpreted.

Most data-driven trajectory prediction systems are formulated as supervised regression problems, where historical aircraft states are mapped to future positions or displacements. Continuous trajectory targets are commonly optimized using regression losses such as Mean Squared Error (MSE), which provides a differentiable training objective and penalizes larger residuals more strongly than smaller ones.

Modern neural trajectory prediction models are typically trained using mini-batch gradient-based optimization. The Adam optimizer introduced by Kingma and Ba (2015) is widely used because it combines adaptive learning rates with estimates of first- and second-order gradient moments. Although Adam is commonly adopted in deep-learning applications, learning rate, batch size, number of epochs, and related settings remain task-specific hyperparameters rather than universally optimal choices.

Regularization is also important because recurrent neural networks may overfit highly correlated trajectory samples. Dropout, introduced by Srivastava et al. (2014), is one common approach, while validation-based early stopping is frequently used to terminate training when further optimization no longer improves generalization performance. These techniques address different aspects of overfitting and are therefore often used in combination.

Horizon-specific evaluation

Trajectory prediction difficulty generally increases with forecast horizon. At shorter horizons, recent velocity, heading, and vertical motion remain highly informative, whereas longer horizons allow more opportunity for maneuvers, route deviations, weather effects, or ATC interventions to alter future motion.

Performance should therefore be assessed separately for different horizons rather than summarized into a single aggregate error. Aircraft trajectory studies vary considerably

in the temporal intervals they examine, and terms such as short-term and medium-term are not used uniformly across the literature. Horizon-specific reporting consequently provides a clearer interpretation of predictive performance.

The present study follows this principle by constructing and evaluating separate horizon-specific models, allowing prediction errors at 5, 10, and 15 minutes to be interpreted independently.

Evaluation in physical units

Training loss alone does not provide a complete measure of trajectory prediction quality. Model outputs should ultimately be evaluated in physically meaningful units after any normalization or target scaling has been reversed.

For horizontal position, great-circle distance provides an appropriate geographic error measure because latitude and longitude are defined on the Earth's surface. Haversine distance is therefore commonly used to express horizontal prediction error in kilometers. Vertical performance should be evaluated separately, since altitude error and horizontal position error represent different aspects of aircraft motion.

A single summary statistic is also insufficient to describe prediction performance. The mean reflects average error but is sensitive to large deviations, while the median provides a more robust measure of typical performance. Upper-percentile measures such as the 90th percentile (p90) provide additional information about the tail of the error distribution and are useful for identifying less favorable prediction cases.

Direct comparison of reported error values across studies should nevertheless be approached cautiously. Differences in geographical region, flight phase, sampling frequency, preprocessing, available features, prediction horizon, target representation, and test-set construction can substantially affect numerical performance. Published trajectory prediction results are therefore most meaningful within the context of their own experimental design.

Generalization and dataset partitioning

Evaluation methodology also determines the strength of any generalization claim. Sliding-window datasets frequently contain strongly overlapping samples. If such

windows are randomly distributed across training, validation, and test subsets, closely related samples may appear in more than one partition.

Random window-level splitting can still be useful for model development and controlled architecture comparison, but it does not provide the same evidence as evaluation on temporally, geographically, or operationally separated data. Stronger protocols may separate samples by complete trajectory, aircraft, route, airport, geographical region, or time period.

Schimpf et al. (2023), for example, emphasized generalized aircraft trajectory prediction across broader operational conditions, illustrating the importance of evaluating whether predictive performance transfers beyond a narrowly defined development dataset. Recent work has also adopted chronological train-validation-test separation to reduce direct temporal overlap between development and evaluation periods.

The present thesis uses a random window-level split during model development and therefore does not interpret this internal test set as a strict independent-flight generalization experiment. To provide a stronger robustness assessment, the final selected models are additionally evaluated on a temporally distinct later-day full-day dataset. This temporal separation offers stronger evidence of transfer than the internal random split, while still remaining limited to the geographical region, altitude range, data source, and time period considered in this study.

This distinction is central to the interpretation of the results. Internal validation supports architecture comparison and model selection, whereas the later-day full-day evaluation serves as a separate temporal robustness check rather than proof of universal operational generalization.

The following section extends this discussion by examining richer context-aware prediction systems, ETA-related tasks, and the specific position of the present thesis within the broader trajectory prediction literature.

2.8 Context-Aware Prediction, ETA and Positioning of the Present Thesis

Recent aircraft trajectory prediction research increasingly supplements state histories with additional contextual information in order to reduce uncertainty about future motion. Flight plans, route structure, aircraft performance, meteorological conditions, ATC intent, and surrounding traffic can all provide predictive information that is not directly observable from recent kinematic states alone.

Flight-plan aware methods are one important example of this development. Georgiou et al. (2020) investigated semantic-aware aircraft trajectory prediction using flight plan information, showing how route semantics can constrain the set of plausible future trajectories. Similar context-aware approaches incorporate intent variables, weather information, aircraft-specific characteristics, or operational constraints to improve prediction beyond what can be inferred from surveillance states alone.

Another extension concerns interactions among multiple aircraft. Social-LSTM-type approaches model the influence of neighboring traffic, while more recent attention and Transformer-based systems combine temporal histories with route and contextual information (Xu et al., 2021). These models operate with richer information sets, but they also require greater data integration and increased model complexity.

This distinction is important when comparing such approaches with state-vector-only prediction. A model using flight plans, weather, aircraft performance, and surrounding traffic is solving a different forecasting problem from one based only on ADS-B surveillance histories. Differences in performance therefore reflect both model architecture and information availability.

The present thesis deliberately adopts the latter, lower-information setting. Future aircraft motion is predicted from recent ADS-B state-vector histories and derived kinematic variables, without using flight plans, weather, aircraft type, explicit route semantics, ATC intent, or neighboring-aircraft information. The purpose is therefore not to approximate the maximum performance achievable in a fully informed operational environment, but to evaluate how much predictive information can be extracted from surveillance states alone.

ETA prediction as a related but distinct task

Estimated Time of Arrival prediction is closely related to trajectory prediction but represents a different forecasting problem. Conventional ETA models estimate the time at which an aircraft will reach a defined destination or operational boundary, such as an airport, runway, waypoint, or terminal-area entry point. Such estimates generally depend on remaining route distance, future speed evolution, traffic conditions, weather, and other operational factors.

A trajectory predictor, by contrast, estimates future spatial states. Although trajectory information can support ETA estimation, a fixed horizon future position prediction does not by itself constitute a complete destination-specific ETA model.

Zheng et al. (2023), in *A Data-Light and Trajectory-Based Machine Learning Approach for the Online Prediction of Flight Time of Arrival*, provide an example of destination-related temporal prediction, where remaining trajectory information is used to estimate arrival at a defined operational boundary and subsequent landing time.

The temporal quantity derived in the present thesis should therefore be interpreted more narrowly. It is computed from the relationship between predicted and realized displacement over a fixed forecast horizon and is used as an additional temporal interpretation of trajectory prediction error. It does not estimate arrival at an airport, waypoint, runway, or other destination-specific target.

For this reason, the metric is more accurately described as a derived ETA-style temporal proxy rather than a conventional ETA prediction. This distinction should remain explicit when interpreting the corresponding results.

Positioning of the present thesis

The literature reviewed in this chapter shows a progression from kinematic and performance-based trajectory prediction toward data-driven sequence models and increasingly context-rich deep learning systems. Modern approaches may combine surveillance data with flight plans, weather, route information, geometric representations, multiple aircraft interactions, or attention-based architectures.

Within this broader landscape, the present thesis adopts a comparatively compact modeling framework. It uses OpenSky ADS-B state vectors, derives relative and kinematic features, and applies LSTM-based sequence models to predict three-dimensional aircraft displacement at predefined elapsed-time horizons. The selected models are additionally evaluated on a temporally distinct later-day dataset.

The contribution of the study should not be interpreted as the introduction of a fundamentally new LSTM architecture, nor as the first use of ADS-B, recurrent trajectory prediction, displacement targets, or multi-horizon forecasting. These concepts are already established in the literature.

Instead, the contribution lies in the structured empirical evaluation of a state-vector-only prediction framework under clearly defined methodological constraints. The study examines the effects of feature enrichment, recurrent capacity, prediction horizon, and temporal evaluation while intentionally excluding richer contextual information.

This positioning offers a useful balance between simplicity and predictive capability. State-vector-only models cannot resolve all sources of future uncertainty, particularly those related to intent, weather, procedures, or traffic interactions, but they provide a comparatively reproducible framework based on information that is widely available in ADS-B surveillance data.

The thesis therefore contributes a focused assessment of how far enriched ADS-B state histories can support three-dimensional endpoint prediction using compact LSTM architectures, complemented by a temporally distinct full-day robustness evaluation and a limited ETA-style temporal interpretation of the resulting trajectory forecasts.

Chapter 3: Data Collection and Preprocessing Pipeline

3.1 Data Source and Operational Scope

The empirical basis of this study consists of ADS-B state-vector data obtained from the OpenSky Network, a large-scale crowdsourced air-traffic surveillance infrastructure (Schäfer et al., 2014; Strohmeier et al., 2021). The extracted dataset contains timestamped aircraft state information including geographic position, barometric altitude, ground velocity, vertical rate, and heading. The available data covered the first week of May 2019. Model development was conducted using hourly ADS-B files from May 1, 2019, while the final robustness evaluation was performed on the complete 24-hour dataset of May 2, 2019. This date was selected to represent a typical weekday traffic profile within European continental airspace. Special filtering was applied using a geographic bounding box defined by $35^{\circ} - 55^{\circ}$ N Latitude, $5^{\circ} - 30^{\circ}$ E Longitude. This region encompasses central and southeastern European airspace, including major enroute corridors and dense traffic flows. To focus on relatively stable cruise phase dynamics and reduce variability associated with climb and descent procedures, altitude filtering was applied to retain only observations above Flight Level 200 (FL200 = 20.000 feet). This altitude filtering reduces the contribution of lower altitude terminal operations and focuses the analysis primarily on higher altitude enroute traffic.

The raw data was exported as hourly CSV files. Each hourly file may exceed 1,000,000 rows depending on traffic density. For the full day evaluation, 24 consecutive hourly files were processed independently and subsequently aggregated for performance analysis.

3.2 Data Cleaning and Structural Preparation

ADS-B data streams may contain inconsistencies, missing fields, and variable sampling intervals. Therefore, a structured preprocessing pipeline was implemented prior to data set construction. Initially, type conversion was applied to ensure consistent numerical formatting of core variables. Records containing invalid or missing values for latitude, longitude, altitude, velocity, vertical rate, or heading were removed. Observations were then sorted by aircraft identifier and timestamp to reconstruct temporally ordered trajectories. Because ADS-B transmissions are not strictly periodic, temporal discontinuities may occur. To prevent unrealistic motion interpolation across

large time gaps, trajectory segmentation was performed. A new trajectory segment was created whenever the time difference between consecutive observations exceeded 60 seconds:

$$\Delta_t = t_i - t_{i-1} > 60 \text{ seconds} \quad (3.1)$$

Each segment portion was assigned a unique trajectory identifier (traj_id). Trajectories shorter than a predefined minimum length were excluded to ensure sufficient historical context for sliding window construction. This segmentation strategy prevents spurious learning from artificially connected trajectory fragments and improves temporal consistency in sequence modeling.

3.3 Feature Engineering and Motion Representation

To improve learning stability and reduce bias associated with absolute geographic coordinates, a delta-based motion representation was adopted. Instead of feeding raw latitude and longitude values to the neural network, first-order differences were computed:

$$\Delta lat_t = lat_t - lat_{t-1} \quad (3.2), \quad \Delta lon_t = lon_t - lon_{t-1} \quad (3.3),$$

$$\Delta alt_t = alt_t - alt_{t-1} \quad (3.4).$$

This representation emphasizes local motion dynamics and reduces the direct dependence of the prediction target on absolute geographic coordinates. Additional kinematic enrichment was incorporated. Ground velocity and vertical rate were retained as continuous variables while heading angle was encoded using sine and cosine transformations:

$$heading_{sin} = \sin(y) \quad (3.5), \quad heading_{cos} = \cos(y) \quad (3.6).$$

This transformation avoids angular discontinuity at $0^\circ/360^\circ$ and enables smooth gradient propagation during training. Therefore, the final input feature vector at each time step includes horizontal displacement components, vertical displacement, ground speed, vertical rate, trigonometric heading representation.

3.4 Sliding Window Dataset Construction

The supervised learning data set was constructed using a fixed length sliding window mechanism. For each trajectory segment, a window of length $L = 120$ time steps, meaning 120 consecutive historical observations, was extracted forming the input sequence:

$$X_t = \{x_{t-119}, \dots, x_t\} \quad (3.7).$$

For each input window ending at timestamp t , the target was defined as the three-dimensional displacement between the current aircraft state and a future state corresponding to a prediction horizon

$$H: y_{t,H} = p_{t_H} - p_t \quad (3.8).$$

where p_t denotes the current aircraft position and p_{t_H} denotes the selected future aircraft position. Since ADS-B observations are not uniformly sampled, t_H was identified using the observation timestamps rather than a fixed row-index offset. Specifically, the first subsequent observation at or after the target time $t + H$ was selected, provided that:

$$| (t_H - t) - H | \leq 10s \quad (3.9)$$

Samples for which no suitable future observation was available within this temporal tolerance were excluded. Separate horizon-specific datasets were constructed for $H \in \{300, 600, 900\}$ seconds, corresponding to prediction intervals of 5, 10 and 15 minutes. This horizon-specific dataset construction ensures that every model is trained using targets corresponding to the actual elapsed forecasting interval rather than to an assumed number of consecutive records.

To maintain computational tractability, a maximum number of approximately 200,000 samples was retained for each dataset. In addition, the number of samples extracted from each individual trajectory segment was limited to 200 in order to avoid over-representing long trajectories. This sampling strategy supports a broader representation of aircraft motion patterns and reduces the risk that a small number of long trajectory segments with extended temporal coverage dominate the training data.

3.5 Statistical Characteristics of the Experimental Dataset

The experimental data used in this study were organized according to distinct model development and evaluation stages. Initial baseline modeling, feature-enrichment experiments, and architecture comparisons were conducted using the first hourly ADS-B file of May 1, 2019. Following model selection, the final horizon specific ENRICH_LSTM96 models were retrained using the first two hourly files of May 1, 2019. The final robustness and derived ETA evaluation was then performed on 24 consecutive hourly files labelled as May 2, 2019. The statistical characterization presented in the following subsections focuses primarily on this full-day evaluation dataset, since it represents the largest and most operationally diverse dataset used in the final experimental stage.

The timestamps contained in the 24 evaluation files span the continuous UTC interval from May 1, 2019 at 21:00:00 UTC to May 2, 2019 at 20:59:59 UTC. Therefore, the date labels of the hourly files and the corresponding UTC timestamps should be distinguished when interpreting the temporal coverage of the dataset.

3.5.1 Dataset Scale and Coverage

The full-day evaluation dataset contains a total of 57,227,360 ADS-B observations distributed across 24 hourly files. A total of 5,568 unique aircraft identifiers (icao24) and 16,889 unique non-missing callsigns were observed during this period.

The volume of observations varied substantially between hourly files, reflecting the changing number of simultaneously observed aircraft and the temporal variation in air traffic activity. The number of observations per hourly file ranged from approximately 438 thousand to 3.82 million, with a mean of approximately 2.38 million observations per hour. The corresponding number of unique aircraft averaged approximately 1,470 per hourly file.

Table 3.5.1 Statistical overview of the full-day evaluation dataset

Statistic	Value
Number of hourly files	24
Total ADS-B observations	57,227,360
Unique aircraft (icao24)	5,568

Statistic	Value
Unique non-missing callsigns	16,889
Mean observations per hourly file	2,384,473
Median observations per hourly file	2,685,127
Minimum observations per hourly file	438,032
Maximum observations per hourly file	3,816,441
Mean unique aircraft per hourly file	1,470
Median unique aircraft per hourly file	1,723
UTC timestamp coverage	May 1, 21:00:00 – May 2, 20:59:59

The difference between the number of unique aircraft identifiers and callsigns is expected because an aircraft identifier may be associated with more than one observed callsign during the analyzed period. In terms of missing information, all variables required for trajectory modeling (timestamp, latitude, longitude, barometric altitude, velocity, heading, and vertical rate) were complete in the extracted dataset. Missing values occurred only in the callsign field, accounting for 323,851 records, or approximately 0.57% of all observations. Since callsign was not used as a predictive input feature, these missing values did not affect the construction of the trajectory prediction datasets.

3.5.2 Descriptive Statistics of ADS-B Variables

Table 3.5.2 summarizes the principal statistical characteristics of the numerical ADS-B variables used to describe aircraft position and motion. Exact mean, standard deviation, minimum, and maximum values were computed using all 57,227,360 observations. To provide additional information about the central part of the distributions without loading the complete dataset simultaneously into memory, the median and the 10th and 90th percentiles were estimated from a stratified random sample of 100,000 observations from each hourly file, corresponding to 2.4 million observations in total.

Table 3.5.2. Descriptive statistics of the main ADS-B variables

Variable	Mean	SD	p10	Median	p90	Minimum	Maximum
Latitude (°)	46.44	4.45	—	—	—	35.00	55.00
Barometric altitude (m)	10,491.43	1,513.21	8,351.52	10,972.80	11,887.20	6,103.62	38,648.64
Ground velocity (m/s)	224.25	29.27	205.52	229.11	249.96	0.00	1,565.10
Heading (°)	197.87	100.37	64.68	190.86	321.25	0.00	359.89
Vertical rate (m/s)	-0.04	3.89	-0.98	0.00	0.65	-165.81	165.81

Note: Mean, standard deviation, minimum, and maximum values were calculated from the complete full-day dataset. The p10, median, and p90 statistics for altitude, velocity, heading, and vertical rate were estimated from a stratified sample of 2.4 million observations.

The geographic coordinates reflect the spatial filtering applied during data extraction, with latitude restricted approximately to 35°- 55° N and longitude to 5°- 30° E. Similarly, the minimum barometric altitude of approximately 6,104 m is consistent with the FL200 lower-altitude threshold applied during extraction.

The central distributions of the kinematic variables are considerably narrower than the observed extreme values. For instance, although isolated velocity and vertical-rate observations reach unusually high absolute values, the median ground velocity is approximately 229.1 m/s, with 80% of the sampled observations lying approximately between 205.5 and 250.0 m/s. Similarly, the median vertical rate is zero, while the 10th and 90th percentiles are approximately -0.98 m/s and 0.65 m/s, respectively. These values are consistent with the cruise-oriented scope of the dataset, in which a large proportion of aircraft observations correspond to relatively stable vertical motion.

The extreme raw values reported in Table 3.5.2 should therefore not be interpreted as representative operating conditions. During the enriched-feature preprocessing stage, velocity and vertical-rate values were additionally constrained to the ranges used by the modeling pipeline in order to reduce the influence of extreme observations.

3.5.3 Preprocessing and Trajectory Structure

The extracted ADS-B observations were subsequently processed independently for each hourly file using the trajectory preparation procedure described in Sections 3.2 and 3.3. Since all variables required by the enriched modeling framework were complete, the initial complete-feature filtering stage retained all 57,227,360 observations.

Trajectory segmentation was performed separately within each hourly dataset. A new segment was created when the temporal gap between consecutive observations of the same aircraft exceeded 60 seconds. Across the 24 hourly processing units, this procedure produced a total of 64,802 hourly trajectory segments. Of these, 31,601 segments satisfied the minimum length criterion of at least 300 observations and were retained for subsequent sequence construction.

After the minimum trajectory length requirement was applied, 55,702,608 observations, corresponding to approximately 97.34% of the original extracted observations, remained available. The calculation of first-order positional differences removed the first observation from each retained trajectory segment, resulting in a final processed dataset of 55,671,007 observations, or approximately 97.28% of the original dataset.

Table 3.5.3. Preprocessing and trajectory statistics

Statistic	Value
Initial extracted observations	57,227,360
Observations after complete-feature filtering	57,227,360 (100.00%)
Observations after minimum trajectory-length filtering	55,702,608 (97.34%)
Observations after delta-feature construction	55,671,007 (97.28%)
Total hourly trajectory segments	64,802
Valid hourly trajectory segments (≥ 300 observations)	31,601
Median sampling interval	1 s
Median hourly p90 sampling interval	1 s
Mean valid segment length	1,762.68 observations
Maximum observed segment length	3,600 observations

The temporal characteristics of the data indicate a high observation frequency. Both the median sampling interval and the median hourly 90th percentile of the sampling interval were 1 second. Nevertheless, the model input sequence was defined by a fixed number of 120 consecutive observations, rather than by a fixed historical duration. The minimum trajectory length criterion was similarly defined as 300 observations, rather than as an assumed number of elapsed seconds.

It should also be noted that trajectory segmentation during the full-day evaluation was performed independently for each hourly file. Consequently, the segment counts reported above represent the sum of the segments identified within the 24 separate hourly processing units and should not be interpreted as the number of continuous trajectories obtained by segmenting a single combined 24-hour dataframe.

3.5.4 Horizon-Specific Evaluation Samples

Following trajectory preprocessing, prediction samples were generated using the time-based sliding-window procedure described in Section 3.4. The final full-day evaluation focused on the 10 and 15-minute prediction horizons, corresponding to the horizon-specific ENRICH_LSTM96 models selected for the final robustness and derived ETA experiments.

Across the 24 hourly files, the 10-minute horizon produced a total of 3,030,963 valid prediction samples, while the 15-minute horizon produced 2,199,177 samples. The lower sample count for the longer forecasting interval is expected because a longer period of valid future trajectory information is required for a candidate input window to produce an acceptable target observation.

Table 3.5.4. Horizon-specific prediction samples in the full-day evaluation

Prediction horizon	Hourly files	Total samples	Mean samples/hour	Minimum/hour	Maximum/hour
10 minutes	24	3,030,963	126,290	22,532	200,000
15 minutes	24	2,199,177	91,632	17,143	150,168

The number of candidate samples extracted from each trajectory segment was limited to 200, while a maximum of 200,000 prediction samples per hourly horizon-specific dataset was retained. Therefore, the maximum value of 200,000 samples observed for

the 10-minute horizon reflects the predefined computational cap rather than the absence of additional valid candidate windows in that hourly file.

Overall, the statistical characteristics demonstrate that the experimental dataset combines a large number of high-frequency ADS-B observations with substantial aircraft and trajectory diversity. At the same time, the preprocessing procedure retains more than 97% of the extracted observations while enforcing temporal continuity and minimum trajectory length requirements. The resulting horizon-specific datasets therefore provide a large-scale empirical basis for evaluating aircraft trajectory prediction under the full-day operational conditions considered in this study.

Chapter 4: Model Architecture and Training Strategy

4.1 Recurrent Modeling Rationale

The trajectory prediction problem addressed in this study is inherently sequential, since future aircraft motion depends on the temporal evolution of preceding states. LSTM-based sequence modeling has previously been applied directly to aircraft trajectory prediction using surveillance data (Shi et al., 2018). Aircraft motion evolves over time according to temporal dependencies that cannot be captured adequately by static regression models. Aircraft trajectories are influenced not only by positional changes but also by instantaneous kinematic properties that describe the aircraft's motion state. Variables such as ground speed, heading and vertical rate provide direct information about how the aircraft is currently moving and in which direction its trajectory is evolving. Incorporating these variables into the input representation provides additional contextual information that cannot always be inferred reliably from positional differences alone. The prediction of displacement at a future horizon requires the model to extract latent motion patterns embedded within historical state transitions.

Recurrent Neural Networks (RNNs) are specifically designed to process ordered sequences by maintaining an internal hidden state that evolves over time. However, classical RNNs suffer from vanishing and exploding gradient problems when modeling long temporal dependencies. To mitigate these limitations, Long Short-Term Memory (LSTM) networks were selected as the core modeling architecture. LSTM networks incorporate gated mechanisms that regulate information flow through memory cells (Hochreiter & Schmidhuber, 1997). These gates allow the model to retain relevant historical motion patterns while discarding irrelevant fluctuations. In the context of aircraft trajectory prediction, this enables the modelling of both short-term kinematic behavior and medium-term motion stability during cruise phase.

4.2 ENRICH LSTM Architecture Family

Three LSTM network architectures were developed and evaluated within the scope of this study:

- ENRICH_LSTM64
- ENRICH_LSTM96
- ENRICH_LSTM64x32 with Dropout(0.2)

All models follow a sequence-to-one regression paradigm. Given an input tensor of shape (L, F) , where $L = 120$ is the window length and F denotes the number of engineered features, the network produces a 3D displacement vector corresponding to the prediction horizon H . The architecture differences lie in depth and width of the recurrent layers.

4.2.1 ENRICH_LSTM64

This configuration employs a single LSTM layer with 64 hidden units. The resulting recurrent representation is passed through two fully connected layers with 64 and 32 units and ReLU activation, respectively, followed by a three-unit linear output layer that predicts the displacement vector. This model serves as a baseline recurrent architecture with moderate capacity.

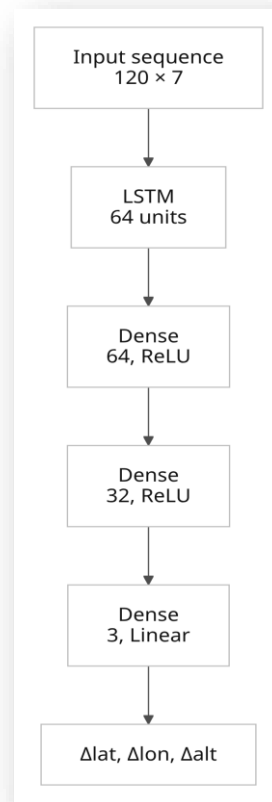


Figure 4.2.1. Architecture of the ENRICH_LSTM64 model. A 64-unit LSTM encodes the enriched historical sequence, followed by a two-layer fully connected regression head and a three-dimensional linear displacement output.

4.2.2 ENRICH_LSTM96

The ENRICH_LSTM96 architecture increases the recurrent layer width to 96 hidden units. The increased hidden-state dimensionality provides the model with greater representational capacity for learning temporal motion patterns. Empirical evaluation showed that this configuration provided the most balanced overall performance across the examined prediction horizons and the evaluation metrics. Consequently, it was selected as the final model.

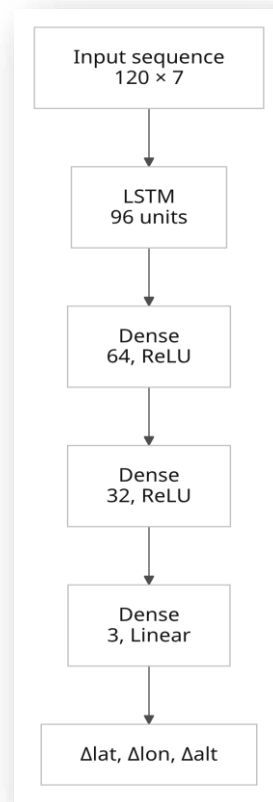


Figure 4.2.2. Architecture of the ENRICH_LSTM96 model. The recurrent representation is increased from 64 to 96 hidden units while the fully connected prediction head is kept unchanged, allowing a controlled assessment of recurrent model capacity.

4.2.3 ENRICH_LSTM64x32 with Dropout

This configuration introduces two stacked LSTM layers with 64 and 32 units respectively. Dropout with a rate of 0.2 is applied after each recurrent layer before the representation is passed to the common Dense(64)→Dense(32) regression head and

the final three-dimensional linear output layer. Stacked recurrent layers increase representational capacity by enabling hierarchical temporal abstraction. However, in the present study, this configuration did not consistently outperform the single layer ENRICH_LSTM96 model.

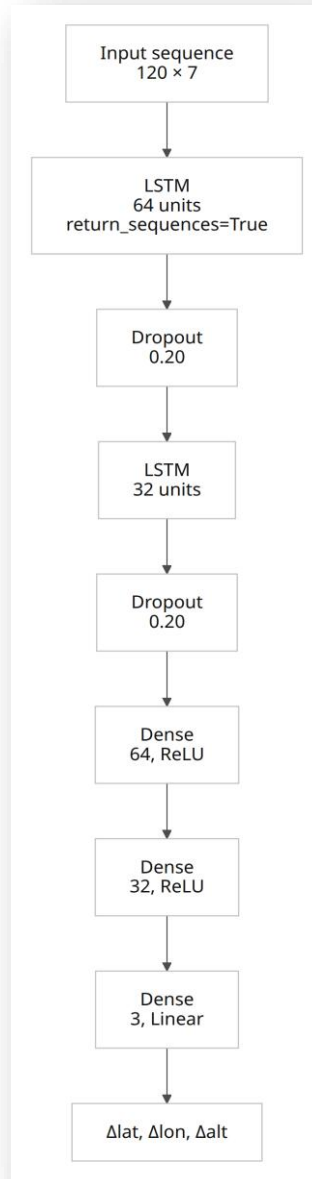


Figure 4.2.3. Architecture of the ENRICH_LSTM64x32_DO20 model. Two recurrent layers with decreasing hidden-state dimensionality are combined with dropout regularization, followed by the same fully connected regression head used by the single-layer variants.

The ENRICH LSTM models evaluated in this study are based on the Long Short-Term Memory framework introduced by Hochreiter and Schmidhuber (1997), which has also

been applied to aircraft trajectory prediction using ADS-B-derived trajectory sequences (Shi et al., 2018). The ENRICH_LSTM64, ENRICH_LSTM96, and ENRICH_LSTM64x32_DO20 architectures were defined as study-specific comparative configurations designed to examine the effects of recurrent hidden-state capacity, recurrent depth, and dropout regularization under a common feature representation and training protocol.

The comparison between 64 and 96 recurrent units should therefore be interpreted as a controlled model-capacity experiment rather than as the reproduction of architectures prescribed by previous work. Similarly, the stacked ENRICH_LSTM64x32_DO20 configuration represents an empirical extension using multiple recurrent layers and dropout regularization. Dropout is an established neural-network regularization technique (Srivastava et al., 2014), while the exact recurrent widths, layer arrangement, and Dense(64)→Dense(32) regression head used here were selected specifically for the experimental comparison conducted in this study.

4.3 Optimization and Training objective

All models were trained as supervised regression networks using the Mean Squared Error (MSE) loss function:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N \|y^{(i)} - \hat{y}^{(i)}\|^2 \quad (4.1),$$

where $y^{(i)}$ denotes the true displacement vector and $\hat{y}^{(i)}$ the predicted displacement for sample i .

MSE was selected because it penalizes larger prediction errors more heavily and provides smooth gradients for stable optimization. It also aligns with standard regression evaluation practices in trajectory prediction literature. Model parameters were optimized using stochastic gradient descent with adaptive learning rate control (Adam optimizer). Mini-batch training was employed to balance computational efficiency and gradient stability. Before model training, both the input features and the displacement targets were standardized using statistics estimated exclusively from the training subset. Each input feature was transformed to zero mean and unit variance, while the three target displacement components were standardized independently. The

same fitted transformations were subsequently applied to the validation and test subsets. During evaluation, predicted displacement vectors were transformed back to their original physical scale before reconstructing the absolute aircraft positions and computing geographic and altitude errors.

4.4 Regularization and Generalization Strategy

To reduce overfitting and support the assessment of model robustness, several regularization strategies were incorporated:

- Delta-based input representation to reduce spatial bias.
- Maximum sample capping and per-trajectory sample limits to reduce the dominance of long or highly represented trajectory segments.
- Dropout (where applicable) to reduce co-adaptation of recurrent units.
- Independent evaluation across 24 hourly files.

Importantly, robustness was not assessed solely on randomly shuffled validation splits. Instead, model stability was evaluated across temporally distinct hourly partitions representing varying traffic densities. This additional evaluation provides a temporally distinct robustness assessment beyond the randomly partitioned development samples.

4.5 Multi-Horizon Training Strategy

Separate models were trained for each prediction horizon of 300, 600 and 900 seconds. Each horizon-specific model was trained together with its corresponding input and target scalars. These fitted scalars were retained and reused during the full day evaluation to ensure that every hourly data set was processed using exactly the same transformations as those employed during training. Independent training ensures that each model specializes in capturing temporal dynamics appropriate for its respective forecasting interval. Prediction error is expected to increase as the horizon expands due to uncertainty propagation and nonlinear motion evolution. Evaluating the model across these three horizons enables systematic assessment of short versus medium-term predictive stability.

4.6 Model Selection

Following the comparative evaluation on the development datasets, ENRICH_LSTM96 was selected as the final architecture because it provided the most balanced combination of mean prediction accuracy and p90 robustness across the examined horizons. It achieved the best performance at the 10-minute horizon and remained competitive at the 15-minute horizon. The selected architecture was subsequently retrained separately for the respective horizons and used in the full day robustness and derived ETA evaluations.

The architectural exploration performed in this study follows an incremental modeling philosophy. Rather than introducing highly complex architectures from the outset, the objective was to evaluate whether moderate increases in model capacity and input feature richness could lead to measurable improvements in trajectory prediction performance. This approach allows improvements to remain interpretable and attributable to specific modeling choices rather than to uncontrolled architectural complexity.

Chapter 5: Experimental Design and Evaluation Framework

5.1 Experimental Objectives

The experimental framework was designed to evaluate the predictive accuracy and robustness of the proposed trajectory prediction models under realistic operational conditions. The primary objective of the experiments is to assess how effectively trained models can forecast aircraft motion in three-dimensional space using only historical ADS-B state vector information. The evaluation focuses on short and medium-term trajectory prediction horizons. Three forecasting intervals were considered: 5 minutes, 10 minutes, and 15 minutes. These horizons represent operationally meaningful timescales in enroute air traffic management. Short term predictions are relevant for tactical monitoring and conflict detection, while medium term horizons provide insight into the evolution of aircraft trajectories over longer intervals.

In addition to evaluating prediction accuracy at different horizons, the experimental design aims to assess the robustness of the models under varying traffic conditions. Aircraft trajectories may exhibit diverse motion patterns depending on routing structures, traffic density, and air traffic control interventions. Therefore, the evaluation framework incorporates statistical measures capable of capturing both average prediction accuracy and performance under more challenging trajectory segments.

5.2 Training Protocol

All models were trained using the supervised learning datasets constructed through the sliding window procedure described in the previous chapter. Each sample in the dataset consists of a fixed length sequence, representing a recent aircraft motion and a corresponding target displacement, representing the aircraft's future position relative to its current state. During the initial development phase, baseline modeling, feature enrichment experiments and architectural comparisons were conducted using the first chronologically available hourly ADS-B file, corresponding to May 1, 2019, from 00:00 to 00:59. After the final model architecture had been selected, separate horizon-specific ENRICH_LSTM96 models were retrained using the first two hourly files of May 1, 2019. The final trained models and their corresponding scalers were subsequently

evaluated on all 24 hourly files of May 2, 2019, which formed a temporally distinct full-day evaluation dataset.

The dataset was randomly shuffled and divided into three subsets: a training set (80%), a validation set (10%) and a test set (10%). The training set was used to learn the model parameters, while the validation set was used for model selection and early stopping. The test subset was reserved for performance evaluation within the development stage random window-level split. Feature and target standardization was fitted exclusively on the training subset to avoid information leakage. The fitted Input scaler was then applied and unchanged to the validation and test sequences, while the target scaler was used to transform the training and validation displacement vectors. Model outputs were inverse transformed before evaluation in the original geographic and altitude units.

Training was performed using mini batch gradient descent. Early stopping was applied by monitoring the validation loss during training and terminating the training process when further epochs no longer produce improvement. The model parameters corresponding to the best validation performance were restored and used for final evaluation. All model configurations were trained under identical conditions, including identical dataset splits, optimization settings and stopping criteria. This controlled training setup facilitates comparison of the effects of architecture and feature representation under otherwise consistent experimental conditions.

5.3 Reconstruction of predicted positions

Although the prediction task is formulated in terms of relative displacement, performance evaluation requires comparison between predicted and true aircraft positions in geographic space. Therefore, predicted displacements were transformed back into absolute coordinates during the evaluation stage. For each prediction sample, the predicted displacement vector was added to the aircraft's current geographic position:

$$\hat{P}_{t+H} = P_t + \Delta\hat{P}_{t+H} \quad (5.1)$$

Where P_t represents the aircraft's position at the final step of the input sequence and $\Delta\hat{P}_{t+H}$ denotes the predicted displacement at horizon H .

This reconstruction step produces an estimated future position in absolute latitude, longitude, and altitude. Before reconstruction, the model output was inverse-transformed from the standardized target space to the original displacement scale. The reconstructed coordinates can then be directly compared with the true aircraft state observed at the corresponding future timestamp. The reconstruction procedure allows the model to benefit from the numerical stability of the delta-based representation during training, while still enabling the use of physically interpretable geographic error metrics during evaluation.

5.4 Evaluation Metrics

Model performance was evaluated using both horizontal and vertical prediction error metrics. These metrics quantify the deviation between the predicted future aircraft position and the true position recorded in the ADS-B data set.

5.4.1 Horizontal Position Error

Horizontal prediction accuracy was measured using the great-circle distance between predicted and true positions. The great-circle distance was computed using the Haversine formula which accounts for the curvature of the Earth and provides an accurate estimate of geographic separation between two coordinates (latitude, longitude). The resulting horizontal error is expressed in kilometers.

Let P_{true} , denote the true aircraft position and P_{pred} the predicted position at the prediction horizon. The horizontal error is defined as

$$E_h = d_{Haversine}(P_{true}, P_{pred}) \quad (5.2)$$

Where $d_{Haversine}$ represents the great-circle distance between the two geographic coordinates.

5.4.2 Vertical Position Error

Vertical prediction accuracy was evaluated using the absolute difference between the predicted and true barometric altitude values.

$$E_u = |alt_{true} - alt_{pred}| \quad (5.3)$$

This metric measures altitude deviation independently of horizontal position and provides insight into the model's ability to capture vertical motion dynamics.

5.4.3 Statistical Error Summaries

To capture different aspects of model performance, several statistical summaries of the error distribution were computed. Mean Error provides a measure of the overall average prediction accuracy across all samples. The Median Error represents the typical prediction outcome and is less sensitive to extreme values in the error distribution.

In addition to these central tendency measures, the 90th Percentile Error (p90) was computed. The p90 metric represents the value below which 90% of the prediction errors fall. This metric is particularly important in trajectory prediction tasks, where a small number of challenging trajectories may produce significantly larger errors than typical cases. Metrics based on percentile, therefore provide insight into the robustness of the model under difficult operational conditions and complement the information provided by mean and median error statistics.

5.5 Horizon Evaluation

The evaluation horizons were defined using actual elapsed time between ADS-B timestamps. Consequently, each target observation corresponded to the required forecasting interval within the predefined temporal tolerance, rather than to a fixed number of data set rows. All models were evaluated across 3 prediction horizons: 5 minutes, 10 minutes, and 15 minutes. This kind of evaluation across horizons allows analysis of how prediction uncertainty evolves as the forecasting horizon increases. As the prediction horizon grows, the model must extrapolate aircraft motion further into the future using only historical state information. This inevitably introduces additional uncertainty, particularly when aircraft undergo heading changes, vertical transitions, or air traffic control interventions. As a result, model performance across multiple horizons provides a comprehensive understanding of both the short-term predictive capability of the model and the limits of medium-term trajectory forecasting. By combining horizon evaluation with mean, median, and percentile statistical metrics, the experimental framework offers a balanced assessment of both prediction accuracy and robustness.

Chapter 6: Results and Performance Analysis

6.1 Baseline Trajectory Prediction Performance

The experiments presented in this chapter were initially conducted using the first hourly ADS-B file of May 1, 2019. After the final architecture had been selected, the horizon-specific ENRICH_LSTM96 models were retrained using the first two hourly files of May 1, 2019. The final models were then evaluated on all 24 hourly files of May 2, 2019 to assess their performance on a temporally distinct full-day dataset.

The first stage of the experimental analysis evaluates the performance of the baseline trajectory prediction model trained using the delta-based representation described in the previous chapters. This baseline configuration utilizes only positional displacement information derived from consecutive ADS-B observations and does not include additional kinematic enrichment features. The purpose of this experiment is to establish the reference level of prediction accuracy against which subsequent model improvements can be assessed. The baseline model was evaluated across the three prediction horizons considered in this study: 5 minutes, 10 minutes, and 15 minutes. For each horizon, the future target state was selected according to the corresponding elapsed time, after which prediction errors were computed after reconstructing the predicted aircraft position from the displacement vector and comparing it with the true ADS-B observation at the corresponding future timestamp. Horizontal prediction accuracy was measured using the Haversine distance between predicted and true geographic coordinates expressed in kilometers.

Table 6.1.1 presents the baseline prediction accuracy across the three horizons in terms of mean error, median error, and the 90th percentile error (p90). These metrics collectively characterize the distribution of prediction errors and allow assessment of both typical performance and worst-case scenarios.

Table 6.1.1 Baseline horizontal prediction error summary

Prediction Horizon	Mean Error (km)	Median Error (km)	p90 Error (km)
5 minutes	7.26	3.99	14.38
10 minutes	14.87	9.90	31.28
15 minutes	19.66	12.87	42.33

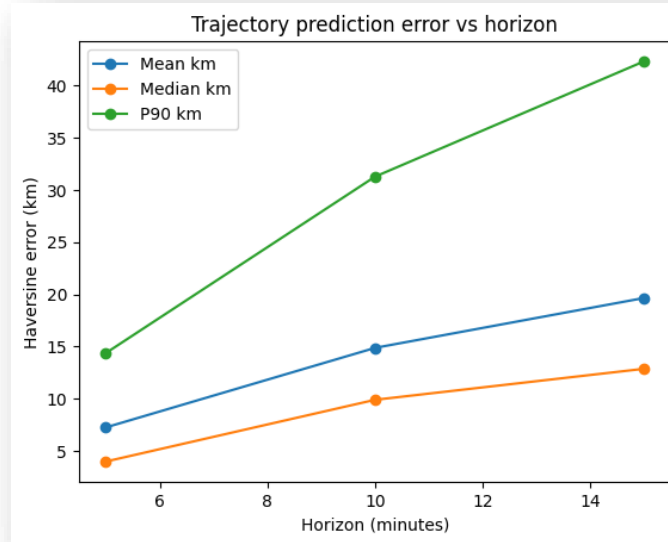
The results show a clear and consistent increase in prediction error as the forecast horizon expands. For the shortest horizon of 5 minutes, the baseline LSTM model achieves a mean horizontal error of 7.26 km with a median error of 3.99 km and a p90 error of 14.38 km. These values establish the baseline performance of the model at the shortest forecasting horizon considered in the study.

When the horizon increases to 10 minutes, the mean prediction error rises to 14.87 km, while the p90 error reaches 31.28 km. Although the prediction error approximately doubles compared with the 5-minute horizon, the increase follows the expected behavior of trajectory forecasting, where uncertainty naturally accumulates over longer prediction intervals.

At the longest horizon of 15 minutes, the mean horizontal error reaches 19.66 km and the p90 error increases to 42.33 kilometers. This gradual degradation in performance reflects the increasing complexity of forecasting aircraft motion over extended time intervals, during which heading changes, altitude adjustments, speed variations, and route modifications become progressively more difficult to anticipate using only historical state-vector information.

The difference between the median and p90 error further illustrates the variability of the prediction task. While the mean errors remain significantly lower than the p90 values, the upper percentile reveals that a subset of trajectories produces substantially larger prediction deviations. These cases typically correspond to trajectories involving maneuvering behavior or deviations from previously observed motion patterns.

Plot 6.1.1, shown below, depicts the relationship between prediction horizon and horizontal trajectory prediction error for the baseline model. A clear monotonic increase in error is observed as the forecasting horizon expands from 5 to 15 minutes, reflecting the expected accumulation of uncertainty over longer prediction intervals. Although the mean and median errors follow similar trends across all horizons, the consistently higher p90 values indicate that a relatively small subset of trajectories exhibits substantially larger prediction errors, resulting in a moderately right-skewed error distribution.



Plot 6.1.1. Baseline horizontal prediction error vs forecasting horizon

As far as vertical prediction errors are concerned, the obtained altitude errors remain moderate across all prediction horizons, reflecting the relatively stable altitude behavior of aircraft during cruise operations above FL200 (Table 6.1.2). Compared with the horizontal component, the vertical prediction accuracy exhibits relatively limited variation across the examined horizons, indicating that altitude changes are generally more predictable than horizontal trajectory evolution during cruise flight.

Table 6.1.2 Baseline vertical prediction error

Horizon	Mean altitude error	Median altitude error
5 min	315.89	167.64
10 min	508.17	261.59
15 min	462.12	184.14

6.2 Impact of Feature Enrichment

After establishing the baseline prediction performance using only positional displacement features, the next stage of the analysis examines the impact of enriching the input representation with additional kinematic information. The baseline model relies exclusively on the positional deltas Δlat , Δlon , Δalt , which describe the recent motion of the aircraft but do not explicitly encode instantaneous motion characteristics. To address this limitation, an enriched feature representation was introduced. In

addition to the positional displacement components, the model input was extended with the following variables.

- Ground velocity,
- Vertical rate,
- Heading represented using sine and cosine transformations.

These features provide direct information about the instantaneous motion state of the aircraft and allow the model to better capture trajectory evolution patterns, particularly when predicting positions over longer horizons.

The enriched model was evaluated using the same experimental setup and evaluation protocol as the baseline configuration. In order to ensure a fair comparison, prediction errors were computed using the same evaluation metrics: the mean horizontal error, median horizontal error and the 90th percentile error (p90). Table 6.2.1 presents the comparison between baseline model and the enriched feature model across the three prediction horizons.

Table 6.2.1. Baseline vs enriched trajectory prediction error

Horizon	Model	Mean Error (km)	Median Error (km)	p90 Error (km)
5 min	Baseline	7.26	3.99	14.38
	Enriched	6.50	3.96	12.75
10 min	Baseline	14.87	9.90	31.28
	Enriched	12.99	8.98	27.44
15 min	Baseline	19.66	12.87	42.33
	Enriched	18.49	13.43	38.75

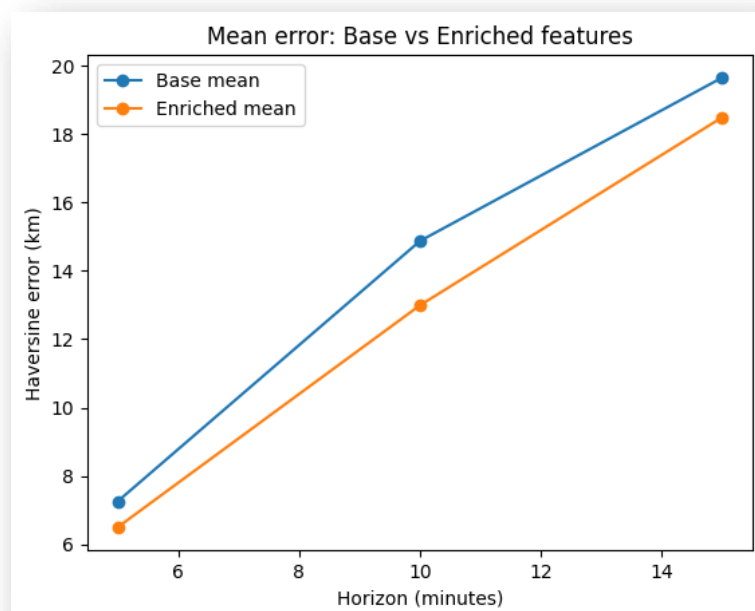
The results indicate that feature enrichment leads to consistent improvements in prediction performance across all horizons. At the first horizon, the mean horizontal error decreases from 7.26 km to 6.5 km, while the p90 error decreases from 14.38 to 12.75 km. Although the improvement is relatively modest, it demonstrates that additional information related to motion helps refine short-term trajectory predictions.

The improvement becomes more pronounced at the 10-minute horizon. Here, the mean error decreases from 14.87 km to 12.99 km, representing a reduction of approximately 1.9 km (12.6%). Similarly, the p90 error decreases from 31.28 km to

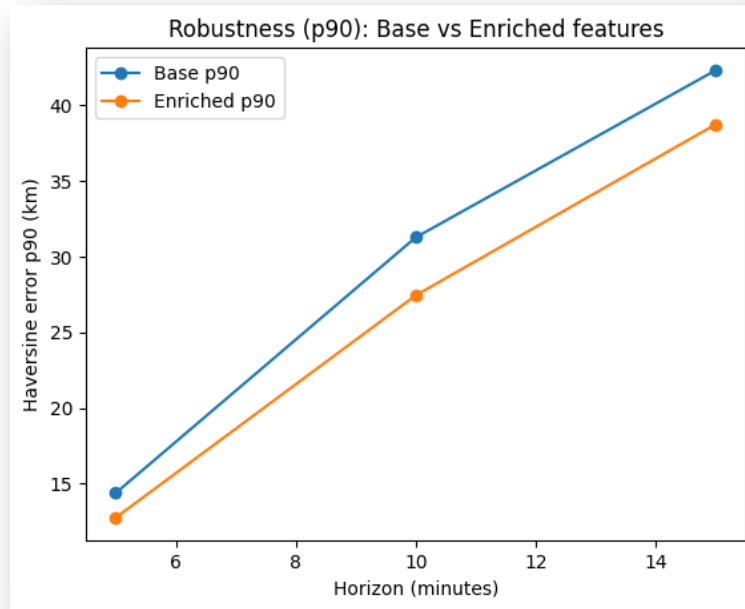
27.44 km, indicating a noticeable reduction in larger prediction deviations. This suggests that the enriched features enable the model to better capture trajectory evolution over medium-term forecasting intervals.

For the longest of three horizons, feature enrichment continues to provide measurable benefits. The mean horizontal error decreases from 19.66 km to 18.49 km, while the p90 error is reduced from 42.33 km to 38.75 km. Although the improvement is smaller than that observed at the 10-minute horizon, the enriched representation consistently improves both average and upper-percentile prediction accuracy, indicating that additional kinematic information remains beneficial even for longer forecasting intervals, despite the naturally increasing uncertainty associated with long range trajectory prediction.

Plotting the results:



Plot 6.2.1. Mean error: Base vs Enriched features



Plot 6.2.2. Robustness (p90): Base vs Enriched features

Plot 6.2.1 illustrates the comparison of mean horizontal prediction error between the baseline and enriched models across the three horizons. The plot shows that feature enrichment consistently improves the mean prediction accuracy at all horizons, with the most pronounced improvement observed for the interval of 10 minutes. Plot 6.2.2 presents the comparison of the P90 error metric, highlighting the effect of feature enrichment on prediction robustness. The consistently lower p90 values across all prediction horizons indicate that the enriched feature representation reduces larger prediction deviations and improves the model's robustness under upper-tail prediction cases, where aircraft motion becomes difficult to predict accurately.

Overall, these results demonstrate that incorporate kinematic information improves the model's ability to capture aircraft motion dynamics. The observed improvements are consistent across all prediction horizons, with the largest relative gain obtained for the 10-minute forecast interval. Although the reductions in mean prediction error are moderate, the systematic decrease in the p90 error indicates that feature enrichment results in more robust and stable predictions by reducing larger prediction deviations that arise in more complex flight scenarios.

In addition to horizontal trajectory prediction accuracy, vertical prediction performance was also evaluated using the absolute altitude error. Table 6.2.2 summarizes the comparison between the baseline and enriched models for each forecasting horizon.

Table 6.2.2. Vertical prediction error comparison

Horizon	Model	Mean Altitude Error (m)	Median Altitude Error (m)
5 min	Baseline	315.89	167.64
	Enriched	274.48	107.77
10 min	Baseline	508.17	261.59
	Enriched	441.13	186.54
15 min	Baseline	462.12	184.14
	Enriched	416.79	168.20

Regarding vertical prediction performance, the enriched feature representation consistently reduces altitude prediction errors across all examined prediction horizons. This behavior is expected given that the dataset focuses on flight segments above FL200, where aircraft generally maintain relatively stable cruise altitudes. The inclusion of additional kinematic information, including vertical rate, provides complementary information that improves the prediction of altitude evolution.

The largest improvement is observed at the 10-minute prediction horizon, where the mean altitude error decreases from 508.17 m to 441.13 m. Similar improvements are also obtained for the two alternate horizons, indicating that the enriched representation contributes to more accurate altitude prediction throughout the examined forecasting intervals.

Consequently, altitude prediction errors exhibit smaller variability across the LSTM architectures under examination in the following subsection. As a result, altitude metrics did not provide additional insight for architecture selection and were not used as part of the model comparison criteria.

6.3 Comparison of ENRICH LSTM Architectures

After establishing the benefit of feature enrichment, the next step of the analysis focuses on evaluating alternative neural network architectures using the enriched input representation. The goal of this stage is to identify the most suitable LSTM configuration for trajectory prediction under the selected experimental setup. As mentioned in subsection 4.2, the following three LSTM architectures were evaluated: ENRICH_LSTM64, ENRICH_LSTM96 and ENRICH_LSTM64x32 with Dropout (0.2).

All architectures were trained using the enriched feature set described in the previous section and evaluated using the same dataset construction procedure and prediction targets. The comparison focuses on the prediction horizons of 10 minutes and 15 minutes. The 5-minute horizon was intentionally excluded from this architectural comparison because it primarily served as a baseline sanity check during the development phase. At very short horizons, architectural differences tend to have a limited impact on performance. In contrast, medium-term horizons such as 10 and 15 minutes present a more challenging prediction problem and therefore provide a more informative setting for comparing model architectures.

To evaluate the models, two performance metrics were used: Mean horizontal error and p90 horizontal error. The mean error measures the average predictive performance of the model across all trajectories, providing an overall indicator of model accuracy. However, trajectory prediction problems often exhibit heavy tailed error distributions due to operational phenomena and changes in aircraft flight dynamics. For this reason, the p90 error was also used as a robustness metric that captures large prediction deviations.

Although the median error was also computed during the experiments, it was not used as a criterion for architecture selection. Since the objective of this comparison is to evaluate both overall predictive accuracy and robustness, the combination of mean error and p90 error provides a more informative assessment of model performance than the median alone. Table 6.3.1 presents the comparative performance of the evaluated architectures.

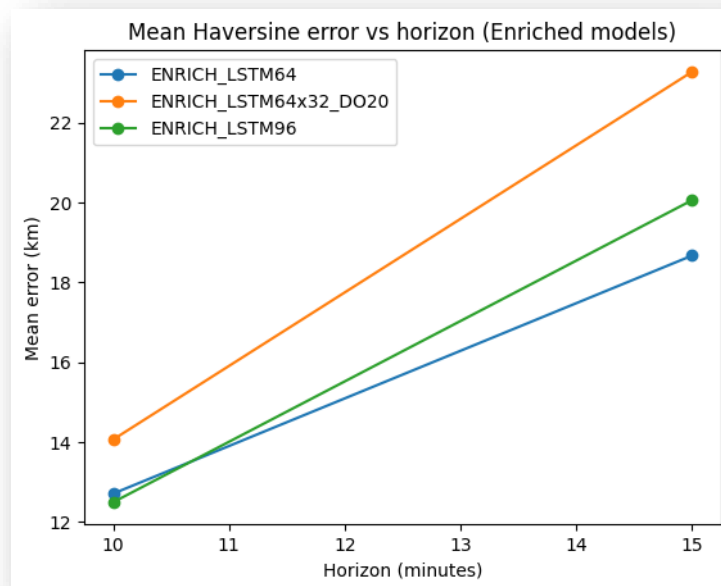
Table 6.3.1. Comparison of ENRICH LSTM architectures

Horizon	Model	Mean Error (km)	p90 Error (km)
10 min	ENRICH_LSTM64	12.70	27.09
	ENRICH_LSTM64x32_DO20	14.06	26.33
	ENRICH_LSTM96	12.49	25.86
15 min	ENRICH_LSTM64	18.67	40.11
	ENRICH_LSTM64x32_DO20	23.28	42.65
	ENRICH_LSTM96	20.06	40.78

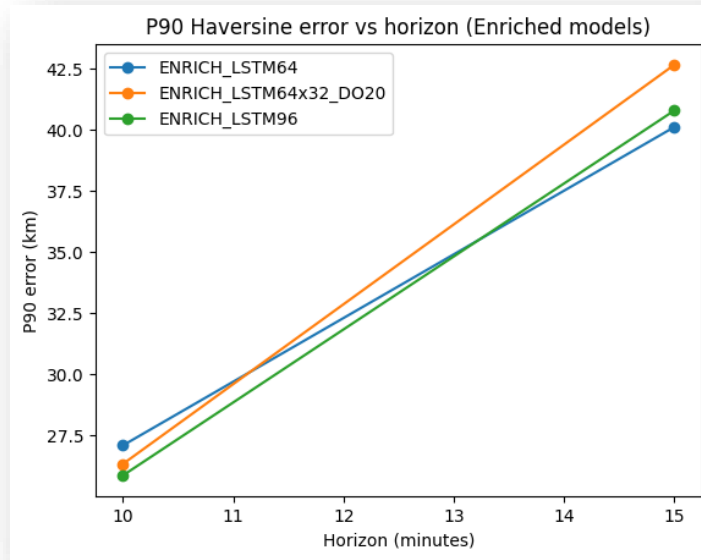
Plotting the results, Plot 6.3.1 illustrates the comparison of mean horizontal prediction error across the evaluated architectures. At the first horizon of 10 minutes the ENRICH_LSTM96 architecture achieves the lowest mean error (12.49 km), followed closely by ENRICH_LSTM64 (12.70 km), while the deeper ENRICH_LSTM64x32_DO20 architecture exhibits the highest mean error (14.06 km). At 15 minutes, the ENRICH_LSTM64 architecture produces the lowest mean error of 18.67 km, whereas ENRICH_LSTM96 follows with a slightly higher value (20.06

km). The ENRICH_LSTM64x32_DO20 architecture again records the highest prediction error, suggesting that the additional architectural complexity does not translate into improved trajectory prediction performance under the examined experimental setup.

Plot 6.3.2 presents the comparison of the p90 error metric across architectures, providing insight into model robustness. Considering the 10-minute horizon ENRICH_LSTM96 achieves the lowest p90 error at 25.86 km, indicating superior robustness compared with the other architectures. At the second horizon of 15 minutes, the ENRICH_LSTM64 architecture achieves the lowest p90 error of 40.11 km, with ENRICH_LSTM96 producing a very similar value (40.78 km). In contrast, the deeper LSTM64x32 architecture again produces the largest p90 error, indicating that the deeper architecture does not improve robustness despite its increased complexity.

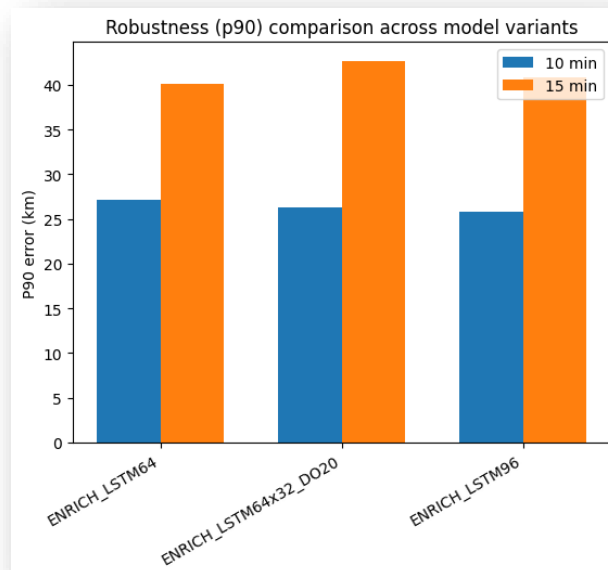


Plot 6.3.1. Mean Haversine error vs horizon (Enriched models)



Plot 6.3.2. *p90 Haversine error vs horizon (Enriched models)*

The robustness comparison within architectures is further depicted on Plot 6.3.3. The bar chart highlights that the LSTM64x32 consistently produces the highest p90 errors at both prediction horizons, whereas the two alternate architectures exhibit very similar robustness, particularly at the 15-minute horizon. These results comply with the indication that the additional architectural complexity of the stacked LSTM configuration does not translate into improved robustness under the experimental conditions considered in this study.



Plot 6.3.3. *Robustness (p90) comparison across model variants*

6.3.1 Model Selection

Considering both evaluation metrics, the ENRICH_LSTM96 architecture was selected as the final model for the remainder of the experiments. Although it does not always produce the lowest mean error, it provides the most balanced overall performance across the evaluated horizons and records the lowest p90 error at the 10-minute horizon, while remaining competitive with ENRICH_LSTM64 at the 15-minute horizon. These results indicate that, under the examined experimental setup, ENRICH_LSTM96 provides the best balance between predictive accuracy and robustness, making it the most suitable architecture for the subsequent full-day evaluation.

Table 6.3.2 summarizes the main experimental stages of the model development process. The results mark the progressive refinement of the trajectory prediction framework. Starting from the baseline displacement representation, the introduction of enriched kinematic features consistently improves prediction accuracy and robustness across the evaluated horizons. The subsequent architectural comparison further assesses alternative LSTM configurations, ultimately leading to the selection of the ENRICH_LSTM96 model as the final architecture used in the full day evaluation presented in the following section.

Table 6.3.2. Summary of the trajectory prediction model development process

Experiment Stage	Model	Horizon	Mean Error (km)	p90 Error (km)
Baseline	LSTM (delta features)	10 min	14.87	31.28
Feature Enrichment	Enriched LSTM		12.99	27.44
Architecture Comparison	ENRICH_LSTM96		12.49	25.86
Baseline	LSTM (delta features)	15 min	19.66	42.33
Feature Enrichment	Enriched LSTM		18.49	38.75
Architecture Comparison	ENRICH_LSTM96		20.06	40.78

6.4 Full-Day Robustness Evaluation

After selecting the ENRICH_LSTM96 architecture as the final trajectory prediction model, a large-scale robustness evaluation was conducted using a full day of ADS-B data. While the earlier experiments were performed on a limited subset of hourly datasets during the model development stage, the final evaluation aimed to assess the model's performance under realistic operational conditions. To achieve this, the trained model was applied to all 24 hourly ADS-B datasets collected on May 2, 2019, covering the entire daily traffic cycle. For each hourly file, trajectories were reconstructed and converted into time-based sliding window prediction samples using the same preprocessing and dataset construction procedure employed during model development. The ENRICH_LSTM96 model was then used to generate trajectory predictions and performance metrics were computed independently for each hour. For every hourly dataset, the following metrics were calculated:

- Mean haversine prediction error (km_mean)
- Median haversine prediction error (km_median)
- p90 haversine prediction error (km_p90)
- Mean altitude prediction error (alt_mean_m)
- Median altitude prediction error (alt_median_m)
- Mean absolute ETA error (eta_mean_abs)
- p90 absolute ETA error (eta_p90_abs)
- ETA errors are reported as absolute errors in minutes.

The hourly results were stored and subsequently aggregated in order to obtain overall performance statistics for the full day dataset. This evaluation procedure provides a temporal robustness assessment, as it evaluates the stability of the selected model under varying density and operational conditions throughout an entire day.

The daily summary values reported below were obtained by aggregating the metrics computed independently for each hourly dataset. In particular, the reported p90 values represent the average of the hourly p90 values rather than a single percentile calculated from all full-day prediction errors pooled together.

Table 6.4.1. Aggregated Full-Day Performance

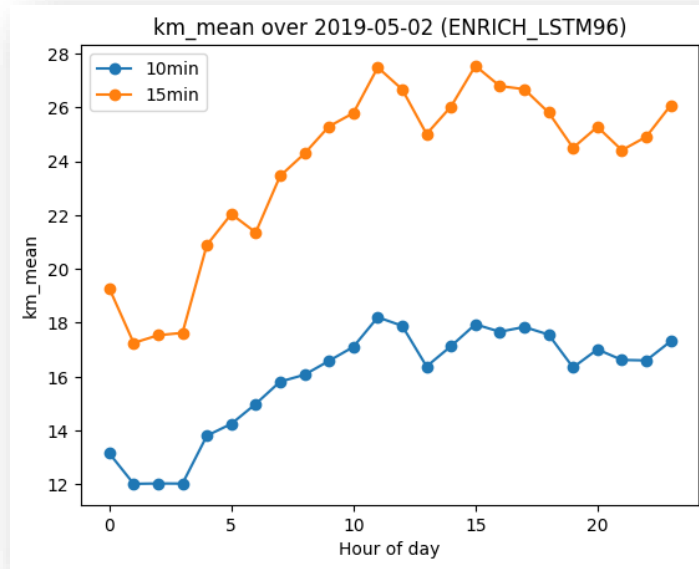
Horizon	Hours	N samples	Mean Error (km)	p90 Error (km)	Mean ETA Error (min)	p90 ETA Error (min)
10 min	24	~3M	15.93	33.79	0.72	1.44
15 min	24	~2.2M	23.83	51.13	0.89	1.90

The results indicate that the selected model maintains stable prediction performance when applied to the complete day of traffic activity. At 10 minutes, the model achieves a mean horizontal prediction error of 15.93 km, while the corresponding p90 error reaches 33.79 km. The associated mean absolute ETA error is only 0.72 minutes, with a p90 absolute ETA error of 1.44 minutes. At the longest horizon, the mean error increases to 23.83 km and the p90 error reaches 51.13 km, reflecting the expected increase in uncertainty over longer forecasting intervals. Despite this increase, the model maintains a mean absolute ETA error below one minute (0.89 min) while the p90 absolute ETA error remains below two minutes (1.90 min) demonstrating stable predictive performance throughout a full day evaluation.

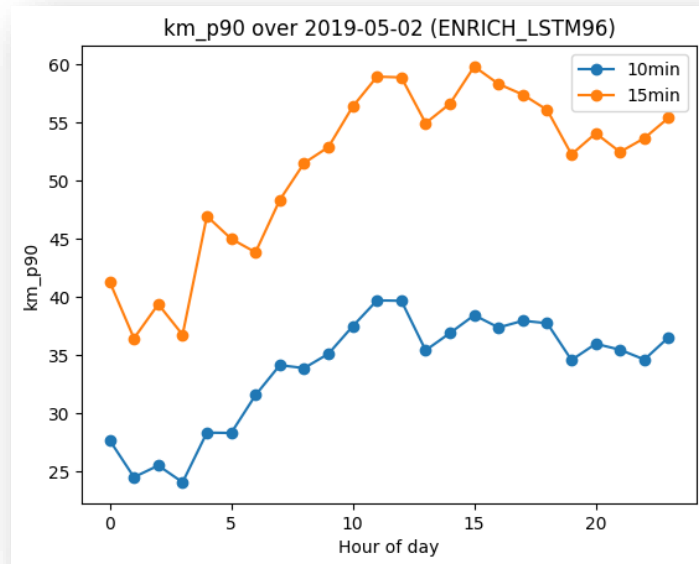
6.4.1 Temporal Stability of Prediction Errors

The hourly evolution of this spatial prediction error is illustrated in Plot 6.4.1. The plot shows that the mean horizontal error remains relatively stable throughout the day for both horizons. For the shortest horizon, the hourly mean error varies approximately between 12 km and 18 km, whereas at the 15-minute horizon it remains within a range of approximately 17 km and 28 km. Although moderate hourly fluctuations are observed, no abrupt deterioration in prediction performance occurs, indicating that the selected model maintains consistent predictive accuracy throughout the daily evaluation period.

Plot 6.4.2 presents the hourly evolution of the p90 spatial error. As expected, the p90 metric exhibits greater variability than the mean error since it reflects the upper tail of the prediction and error distribution. Nevertheless, the p90 values remain within a relatively narrow range throughout the day for both prediction horizons, suggesting that the robustness of the selected model remains stable across different hourly data sets without evidence of systematic performance degradation.



Plot 6.4.1. Hourly spatial error stability



Plot 6.4.2. Robustness over time

6.4.2 Derived ETA Prediction Analysis

In addition to spatial trajectory prediction accuracy, the predicted displacements were used to derive the ETA-style temporal metric introduced in this study. This quantity is based on the

relationship between predicted and realized displacement over the fixed forecasting horizon and should not be interpreted as a conventional destination-specific ETA prediction.

As summarized in Table 6.4.1, the derived ETA-style temporal error remains numerically small across both evaluated horizons. The mean absolute error is 0.72 minutes at the 10-minute horizon and 0.89 minutes at the 15-minute horizon, while the corresponding p90 absolute errors are 1.44 and 1.90 minutes. These values characterize the behavior of the derived fixed-horizon temporal proxy.

The relatively small difference between the mean and p90 ETA errors suggests that the ETA estimation procedure remains stable across the evaluated dataset. Although ETA is derived indirectly from the predicted aircraft trajectory, the obtained results indicate that the proposed trajectory prediction framework provides sufficiently accurate spatial predictions to support reliable ETA estimation under the examined experimental conditions.

Table 6.4.2 summarizes the relationship between spatial prediction error and the resulting ETA error.

Table 6.4.2 Horizon-normalized spatial and ETA prediction errors

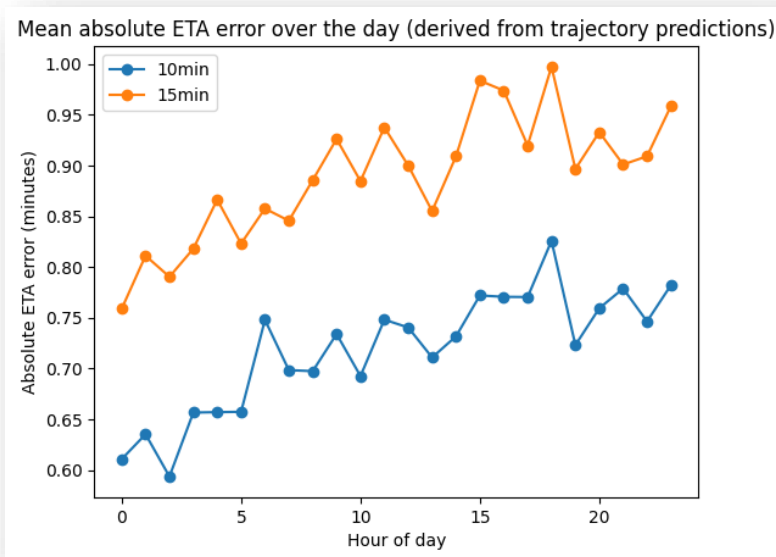
Horizon	Mean Error (km)	Mean ETA Error (min)	Mean Error per minute of horizon (km)	Mean ETA Error per minute of horizon (min)
10 min	15.93	0.72	1.59	0.07
15 min	23.83	0.89	1.59	0.06

The figures show that spatial prediction errors propagate into ETA prediction errors in a relatively consistent manner, showing the same behavior. Despite the increase in prediction horizon from 10 to 15 minutes, the mean ETA error remains below one minute, increasing only from 0.72 to 0.89 minutes. This indicates that the proposed framework maintains reliable temporal prediction accuracy even as spatial uncertainty gradually increases.

To further investigate how prediction errors scale with the forecasting horizon, both the spatial prediction error and the corresponding ETA error were normalized by the prediction horizon. The resulting metrics, expressed as kilometers per minute of prediction horizon and the minutes of ETA error per minute of prediction horizon, provide a normalized measure of error growth. The normalized spatial prediction error remains virtually constant at approximately 1.59 km per minute of prediction horizon for both forecasting intervals, indicating an almost

linear increase in spatial prediction uncertainty as the horizon expands. Similarly, the normalized ETA error remains remarkably stable, decreasing slightly from 0.07 to 0.06 minutes of ETA error per minute of prediction horizon. This behavior suggests that the temporal prediction accuracy scales proportionally with the forecasting horizon and that the ETA estimation procedure remains robust despite the increase in spatial prediction uncertainty.

Plot 6.4.3 illustrates the hourly evolution of the mean ETA prediction error. The results show that the ETA error remains highly stable throughout the day for both prediction horizons. At the 10-minute horizon, the mean absolute ETA error remains below one minute across all hourly datasets, while at the 15-minute horizon it stays within a similarly narrow range, only slightly higher overall. These findings indicate that the derived ETA estimation procedure maintains consistent temporal accuracy throughout the full-day evaluation period.

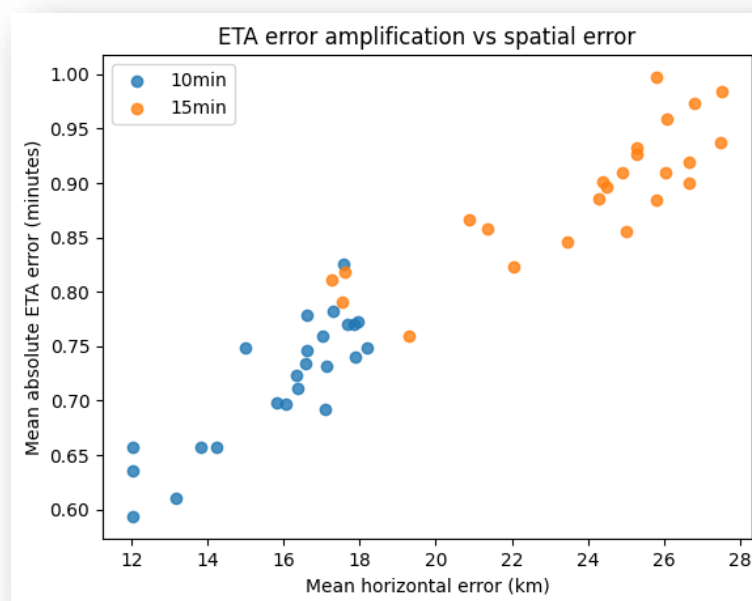


Plot 6.4.3. ETA mean error over the day

The relationship between spatial prediction error and ETA estimation error is further illustrated in plot 6.4.4, which presents a scatter plot of mean horizontal error versus mean absolute ETA error across the evaluated hourly datasets. The plot reveals a clear positive association in the plotted hourly values between spatial and temporal prediction errors, indicating that larger trajectory prediction errors are generally accompanied by larger ETA estimation errors. At the same time, the two prediction horizons form distinct clusters, showing that the forecasting horizon influences the scale of both spatial and temporal prediction uncertainty. For the 10-minute horizon, the mean horizontal error is concentrated at lower values and is associated

with mean absolute ETA errors of approximately 0.6 to 0.8 minutes. For the 15-minute horizon, the horizontal error shifts to higher values with the corresponding mean absolute ETA error remaining below one minute in most cases. This clustering pattern indicates that the increase in ETA error with forecasting horizon is systematic but remains well controlled.

Overall, the results confirm that improvements in spatial trajectory prediction accuracy are directly reflected in improved ETA estimation performance. More importantly, they show that the proposed framework is able to provide reliable derived ETA estimates under realistic full day operational conditions with limited temporal error, even at the longer prediction horizon.



Plot 6.4.4. ETA vs spatial error scatter

Chapter 7: Discussion and Conclusion

7.1 Summary of Key Findings

This study investigated aircraft trajectory prediction in three-dimensional space using ADS-B state vector data and Long Short-Term Memory neural networks. The experimental analysis examined the effect of prediction horizon, input feature representation, recurrent architecture and evaluation scale on model accuracy and robustness.

The baseline model, which relied exclusively on positional displacement features, achieved mean horizontal errors of 7.26 km, 14.87 km and 19.66 km for the 5, 10 and 15-minute prediction horizons, respectively. The corresponding p90 errors increased from 14.38 km at 5 minutes to 31.28 km at 10 minutes and 42.33 km at 15 minutes. These results demonstrate a clear and expected deterioration in predictive accuracy as the forecasting interval expands, while confirming that the adopted time-based target construction produces realistic short and medium-term trajectory estimates.

The introduction of enriched kinematic features, including ground velocity, vertical rate, and sine-cosine heading representation, produced consistent improvements across all examined horizons. The largest relative improvement in mean horizontal error was observed at the 10-minute horizon, where the error decreased from 14.87 km to 12.99 km. Feature enrichment also reduced the p90 error at every horizon, indicating improved robustness against larger prediction deviations. Vertical prediction performance improved consistently as well, although altitude-related metrics exhibited less variation and were not used as primary criteria for architecture selection.

The comparison of alternative enriched LSTM architectures showed that greater architectural complexity did not necessarily lead to better performance. The stacked ENRICH_LSTM64x32_DO20 configuration consistently produced higher mean or p90 errors than the single-layer alternatives. ENRICH_LSTM96 achieved the best mean and p90 performance at the 10-minute horizon and remained competitive at the 15-minute horizon. Considering both average accuracy and upper-percentile robustness, it provided the most balanced overall performance and was therefore selected for the subsequent full-day evaluation.

The full-day robustness experiment showed that the selected horizon-specific ENRICH_LSTM96 models retained broadly consistent performance when applied to a temporally distinct later-day dataset. Across all 24 hourly ADS-B datasets of May 2, 2019, the model achieved a mean horizontal error of 15.93 km and a p90 error of 33.79 km at the 10-minute horizon. At the 15-minute horizon, the corresponding errors were 23.83 km and 51.13 km. Although these values were moderately higher than those obtained during model development, the hourly error profiles remained stable and showed no evidence of systematic performance degradation during the day.

Finally, the derived ETA analysis showed that the predicted trajectory displacements can support stable horizon-based temporal estimates. The mean absolute ETA error remained below one minute for both evaluated horizons, reaching 0.72 minutes at 10 minutes and 0.89 minutes at 15 minutes. The corresponding p90 errors were 1.44 and 1.90 minutes. Moreover, the normalized horizontal error remained approximately constant at 1.59 km per minute of prediction horizon, while the normalized ETA error remained stable at approximately 0.06-0.07 minutes per minute of horizon. These findings indicate that spatial uncertainty increased approximately proportionally with the forecasting interval, without producing disproportionate growth in the derived temporal error.

7.2 Interpretation of Results

The results of this study highlight several important characteristics of data-driven aircraft trajectory prediction. First, prediction accuracy is strongly influenced by the forecasting horizon. As the horizon increases, the model must extrapolate aircraft motion further beyond the observed historical sequence, increasing the possibility that future motion will differ from the recent trajectory pattern. The progressive increase in mean and p90 errors from 5 to 15 minutes is therefore consistent with the increasing uncertainty associated with medium-term trajectory forecasting.

The observed benefit of feature enrichment demonstrates the importance of explicitly representing the aircraft's kinematic state. Positional displacement features describe the recent evolution of the trajectory, while ground velocity, vertical rate, and heading provide complementary information regarding the direction and magnitude of motion. The systematic reductions in mean and p90 horizontal error indicate that these variables improve the information available to the LSTM model. Nevertheless, the experiments do not isolate the

individual contribution of each enriched variable. Therefore, the observed improvement should be attributed to the enriched representation as a whole rather than to any single feature.

The architectural comparison further indicates that increasing neural network complexity does not automatically improve predictive performance. The deeper stacked LSTM configuration produced higher errors than the single-layer alternatives in most of the examined comparisons. This suggests that the relationship between model capacity and trajectory prediction performance depends on the available data, the input representation, and the selected training configuration. Under the present experimental conditions, a moderately sized single-layer LSTM provided a more effective balance between representational capacity and generalization.

The distinction between mean and p90 error also provides important insight into model performance. Mean error summarizes average predictive accuracy, whereas p90 describes the upper region of the observed error distribution. The substantially higher p90 values indicate that a minority of prediction samples produce considerably larger spatial deviations than the typical case. Although these samples may be associated with changes in aircraft motion or other unobserved operational factors, the present analysis does not directly identify their causes. The p90 metric should therefore be interpreted as an indicator of upper-tail prediction performance rather than as direct evidence of specific maneuvering behavior.

The full-day evaluation provides evidence that the selected model retains broadly consistent performance when applied to temporally distinct data from May 2, 2019. The hourly profiles of both mean and p90 error showed moderate fluctuations but no systematic deterioration during the evaluated day. This finding suggests that the model learned motion patterns that remained applicable beyond the limited development subset. However, the results should be interpreted as evidence of temporal generalization within the same geographic region, altitude range, and observation period, rather than as proof of universal generalization across seasons, airspaces, or operational environments.

The normalized error analysis provides further insight into the effect of the forecasting horizon. The mean horizontal error per minute of horizon remained approximately constant at 1.59 km for both the 10 and 15-minute intervals. This indicates that, within the examined range, the growth of average spatial error was approximately proportional to the forecasting horizon. The result does not establish a general linear law beyond the evaluated intervals, but it describes the observed scaling behavior between the two medium-term horizons considered in the full-day experiment.

The derived ETA results followed a similarly controlled pattern. Although larger spatial errors were generally associated with larger temporal errors, the mean absolute ETA error remained below one minute and the p90 absolute error below two minutes at both horizons. Moreover, the normalized ETA error did not increase with the forecasting horizon, decreasing slightly from 0.07 to 0.06 minutes of ETA error per minute of horizon. Consequently, the findings do not indicate disproportionate temporal error amplification. Instead, they indicate that the derived temporal error scaled in a stable manner under the experimental conditions examined.

From an operational perspective, these results suggest that accurate short and medium-term trajectory predictions may support downstream temporal assessments. However, the ETA measure used in this study is derived from displacement over a fixed prediction horizon and should not be interpreted as a complete destination-specific arrival time prediction. Its practical significance lies primarily in demonstrating the relationship between predicted spatial motion and a corresponding horizon-based temporal estimate. More comprehensive operational ETA applications would require destination information, route structure, flight intent, weather conditions, and air traffic management constraints.

7.3 Limitations of the Study

Despite the encouraging results, several limitations of the present study should be acknowledged. First, the proposed framework relies exclusively on historical ADS-B state-vector observations and does not incorporate external contextual information such as weather conditions, flight plans, aircraft type, air traffic control instructions, route structure, or airspace constraints. These factors may substantially affect future aircraft motion, particularly when an aircraft deviates from its recent trajectory pattern. Consequently, the predictive capability of the model is limited to information that can be inferred directly from the observed kinematic sequence.

Second, the prediction framework follows a sequence-to-one formulation. For each historical input window, the model estimates a single three-dimensional displacement at a fixed future horizon rather than predicting the complete intermediate trajectory. Although this formulation simplifies training and enables direct evaluation at operationally relevant horizons, it does not explicitly represent how the aircraft is expected to move between the current and predicted future states. Therefore, the model cannot describe the continuous evolution of the trajectory or identify when an intermediate maneuver may occur.

A further limitation concerns the temporal representation of the historical input. Each input sequence contains 120 consecutive ADS-B observations, but these observations do not necessarily correspond to a fixed duration because ADS-B sampling intervals are not perfectly uniform. The future target horizons were defined using actual elapsed time and a temporal tolerance, whereas the historical context remained observation-count based. As a result, the amount of past time represented by each input sequence may vary between trajectory segments.

The experimental data are also restricted in temporal and geographic scope. Model development was conducted using a limited subset of hourly ADS-B data, while the full-day evaluation was performed on May 2, 2019, within a predefined European geographic region and for observations above FL200. Although the hourly evaluation provides evidence of temporal stability within the examined day, it does not establish generalization across different seasons, years, geographic regions, airspace structures, or traffic regimes. The conclusions should therefore be interpreted within the boundaries of the available dataset. The focus on observations above FL200 represents another important limitation. Aircraft motion during cruise is generally more stable than during climb, descent, approach, or departure. Consequently, the reported results may not transfer directly to terminal-area operations, where aircraft trajectories are influenced by more frequent altitude changes, turns, speed adjustments, and procedural constraints.

During model development, the constructed prediction samples were randomly divided into training, validation, and test subsets. Since multiple overlapping windows may originate from the same trajectory segment, closely related samples could be distributed across different subsets. This may lead to a more optimistic estimate of development-set performance than a split performed strictly by aircraft, trajectory, or time period. The separate full-day evaluation partially addresses this concern by testing the selected models on a temporally distinct day, but it does not eliminate the limitation of the sample-level split used during model comparison.

The architecture comparison was based on the obtained performance from a limited set of model configurations and fixed experimental settings. No systematic hyperparameter optimization, repeated training across multiple random seeds, or formal statistical significance testing was performed. Therefore, relatively small differences between architectures should be interpreted as numerical differences under the specific experimental setup rather than as definitive evidence that one architecture is universally superior. The aggregation of the full-day results also requires careful interpretation. The reported daily summary values were obtained by aggregating metrics computed independently for each hourly dataset. In particular, the

reported p90 value summarizes the hourly p90 metrics and does not correspond to a single percentile calculated from all individual full-day prediction errors pooled together. This approach is appropriate for examining hourly stability, but it should not be interpreted as an exact global full-day error distribution.

Finally, the ETA measure used in this study is analytically derived from predicted and observed displacement over a fixed forecasting horizon. It does not represent a destination-specific Estimated Time of Arrival based on the remaining route to an airport. The calculation does not incorporate destination information, route geometry, flight intent, weather conditions, or operational restrictions, and should therefore be interpreted as a derived horizon-based temporal estimate. Although the obtained errors demonstrate a stable relationship between spatial prediction and temporal estimation, they are not directly comparable with the output of specialized commercial-flight ETA prediction systems.

7.4 Future Work

Several directions for future research can be identified based on the findings and limitations of this study. A primary extension would be the incorporation of additional contextual information beyond ADS-B state vectors. Weather conditions, flight plans, aircraft type, route structure, airspace constraints, and air traffic control information could provide valuable explanatory variables for changes in aircraft motion that cannot be inferred solely from recent kinematic observations. The integration of such features may be particularly beneficial at longer forecasting horizons, where uncertainty regarding future aircraft behavior becomes more pronounced.

Future research should also consider alternative approaches to the temporal representation of the input sequence. In the present framework, each input window contains a fixed number of consecutive ADS-B observations, while the actual elapsed duration represented by these observations may vary because of irregular sampling intervals. A time-normalized approach based on trajectory resampling or interpolation at a fixed temporal frequency could provide a more consistent historical context. Alternatively, models that explicitly incorporate elapsed-time differences as an input feature could be investigated without requiring complete resampling of the original trajectories.

Another important extension concerns the prediction formulation. The sequence-to-one approach used in this study estimates aircraft displacement at a single future horizon. Future

work could adopt sequence-to-sequence or multi-output forecasting methods capable of predicting a complete sequence of intermediate future states. Such approaches would provide a continuous representation of the expected trajectory and enable the evaluation of error accumulation throughout the prediction interval rather than only at its endpoint.

A more extensive exploration of neural-network architectures would also be valuable. Attention-based recurrent models, Temporal Convolutional Networks, Transformer architectures, and hybrid physics-informed machine learning approaches could be compared with the LSTM models examined in this study. However, future architectural comparisons should be conducted using systematic hyperparameter optimization, multiple random seeds, and appropriate statistical procedures in order to determine whether observed performance differences are consistent rather than dependent on a single training run.

The evaluation protocol could also be strengthened through trajectory-level or time-based dataset partitioning. Splitting the data by aircraft, complete trajectory segment, hour, or day would reduce the possibility that overlapping or closely related prediction windows appear in different development subsets. Evaluating models across multiple independent days, seasons, years, and geographic regions would provide a more comprehensive assessment of generalization and make it possible to examine performance under a broader range of traffic and operational conditions.

Further research should extend the analysis beyond cruise operations above FL200. Aircraft trajectories during climb, descent, approach, and departure involve more frequent changes in speed, heading, and altitude and therefore represent a more demanding prediction setting. Developing phase-specific models or incorporating the flight phase as an additional input feature could improve the representation of these distinct motion regimes.

Future studies could additionally examine uncertainty-aware trajectory prediction. Instead of producing only a single deterministic displacement estimate, probabilistic models could generate prediction distributions, confidence regions, or multiple plausible future trajectories. Techniques such as quantile regression, Bayesian neural networks, ensemble methods, or mixture-density models may provide more informative measures of prediction uncertainty for operational decision support.

The ETA component could also be extended beyond the horizon-based temporal estimate used in the present study. A destination-specific ETA framework would require information regarding the destination airport, remaining route, aircraft intent, expected speed profile,

weather conditions, and operational restrictions. Combining trajectory prediction with route-aware or flight-plan-aware ETA estimation would enable a more direct comparison with specialized commercial-flight ETA models and provide greater operational relevance for arrival sequencing and traffic-flow management.

Lastly, future work should consider the practical integration of trajectory prediction models into air traffic management systems. This would require evaluation using streaming ADS-B data, assessment of computational latency, continuous model monitoring, and analysis of model performance under changing operational conditions. Human-factor considerations and the presentation of prediction uncertainty to air traffic controllers or operational decision-makers would also be essential before any real-world deployment.

7.5 Concluding Remarks

This thesis presented a data-driven framework for short and medium-term aircraft trajectory prediction in three-dimensional space using ADS-B state-vector data and Long Short-Term Memory neural networks. The proposed approach combined trajectory preprocessing, time-based target construction, delta-based motion representation, feature enrichment, horizon-specific model training, and evaluation using physically interpretable horizontal and vertical error metrics.

The experimental results showed that prediction accuracy decreases progressively as the forecasting horizon increases, reflecting the greater uncertainty associated with longer-term motion extrapolation. At the same time, the inclusion of kinematic variables such as ground velocity, vertical rate, and heading consistently improved both average horizontal accuracy and upper-tail performance compared with the baseline positional representation. The comparison of alternative recurrent architectures further demonstrated that increased architectural complexity does not necessarily produce better results, and that a moderately sized single-layer LSTM can provide a suitable balance between predictive accuracy and robustness.

A central contribution of the study was the evaluation of the selected model across a complete operational day. The relatively consistent hourly error patterns observed across 24 temporally distinct datasets indicate that the learned trajectory representations remained applicable beyond the limited development subset. Although the evaluation was restricted to a specific geographic region, altitude range, and observation period, the results provide encouraging

evidence that the proposed framework can maintain stable performance under changing hourly traffic conditions.

The derived temporal analysis also showed that the predicted spatial displacements can support consistent horizon-based temporal estimates. Mean absolute temporal errors remained below one minute for both the 10 and 15-minute horizons, while the normalized ETA error did not increase disproportionately as the forecasting interval expanded. These findings indicate that the spatial predictions generated by the model retain useful temporal information. Nevertheless, the derived ETA measure should be interpreted as a fixed-horizon temporal estimate rather than as a complete destination-specific arrival-time prediction.

Overall, the study demonstrates that carefully constructed ADS-B trajectory datasets, appropriate temporal target alignment, and enriched kinematic representations can support effective data-driven aircraft motion forecasting. The findings also emphasize that reliable evaluation requires more than reporting average prediction error. Horizon-dependent analysis, upper-percentile metrics, temporally distinct testing, and explicit recognition of methodological limitations are all necessary for a balanced assessment of trajectory prediction performance. While the proposed framework does not incorporate flight plans, weather information, aircraft intent, or destination-specific operational constraints, it establishes a coherent and reproducible basis for further research. Future extensions incorporating richer contextual information, more rigorous temporal partitioning, probabilistic forecasting, multi-step trajectory generation, and route-aware ETA modeling could build upon the present work and further improve the operational relevance of data-driven trajectory prediction systems.

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Appendix: Code and Reproducibility

The source code, implementation notebook, software requirements, detailed reproduction instructions, and final thesis document are publicly available in the accompanying GitHub repository:

[<https://github.com/antonis00h/Aircraft-trajectory-prediction-adsb>]